

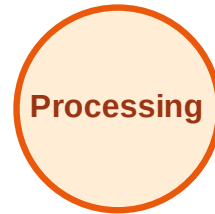
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

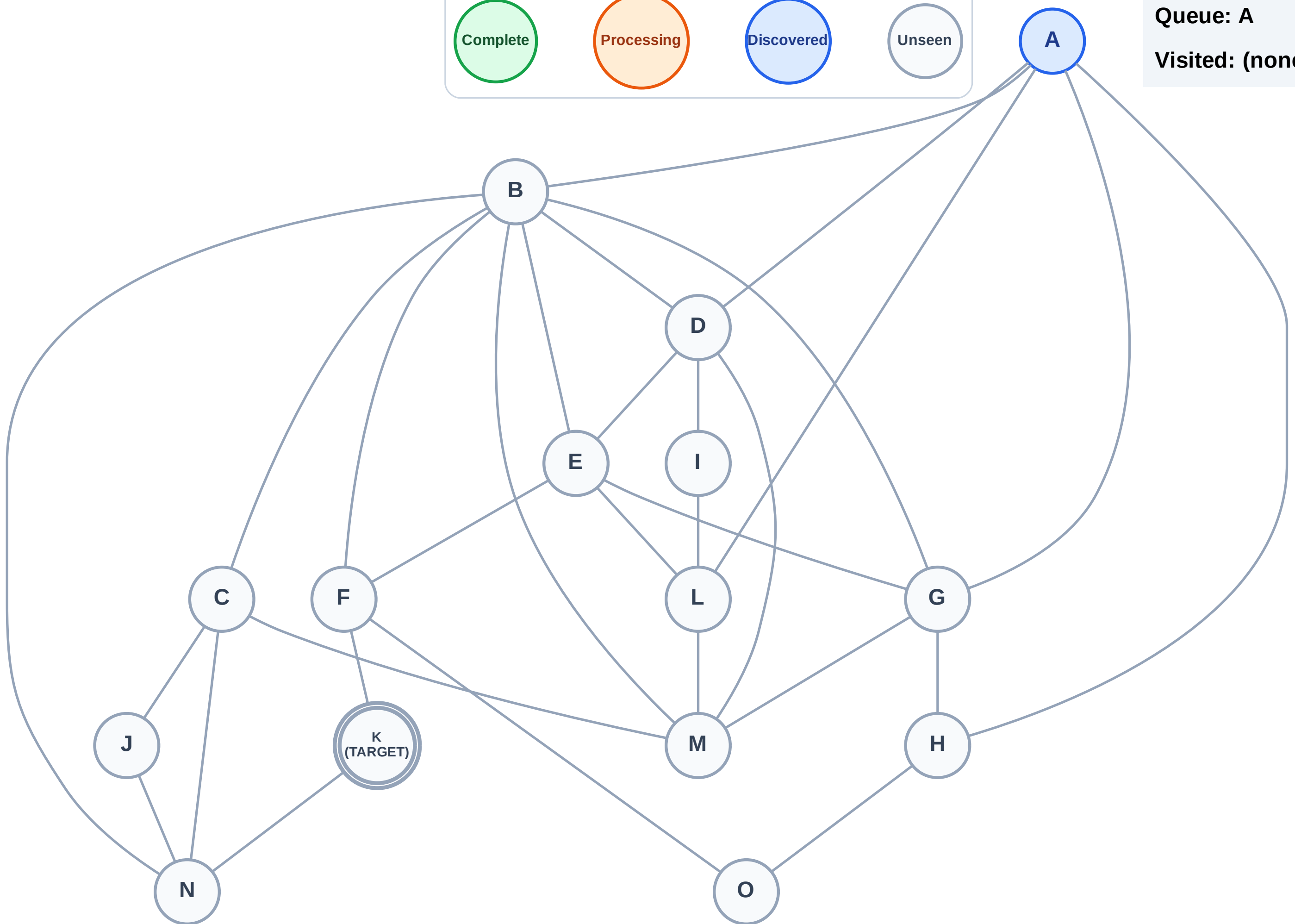


Discovered



Unseen

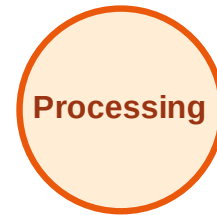
Queue: A
Visited: (none)



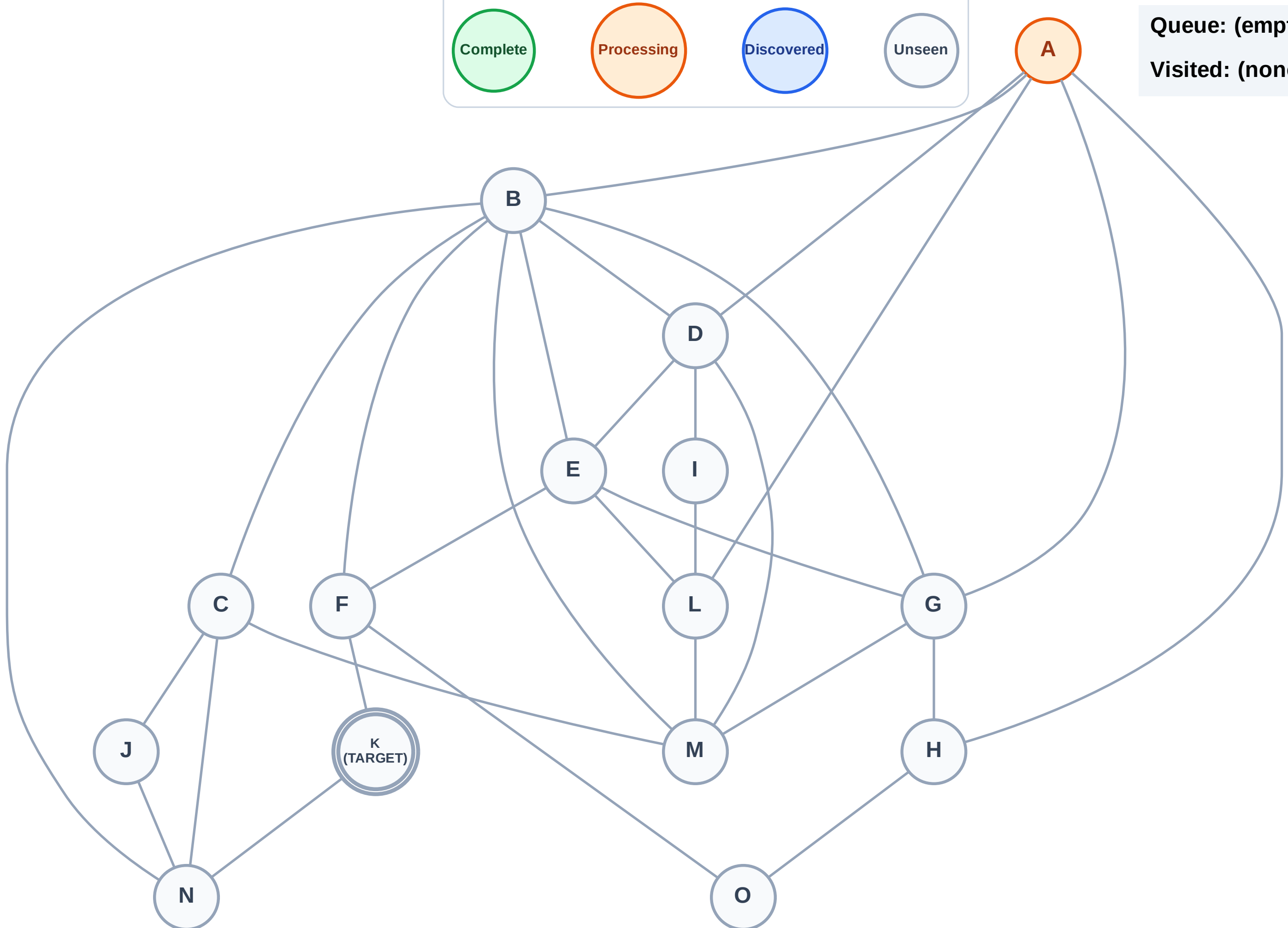
BFS Traversal

Step 2: Process node A

Legend



Queue: (empty)
Visited: (none)



BFS Traversal

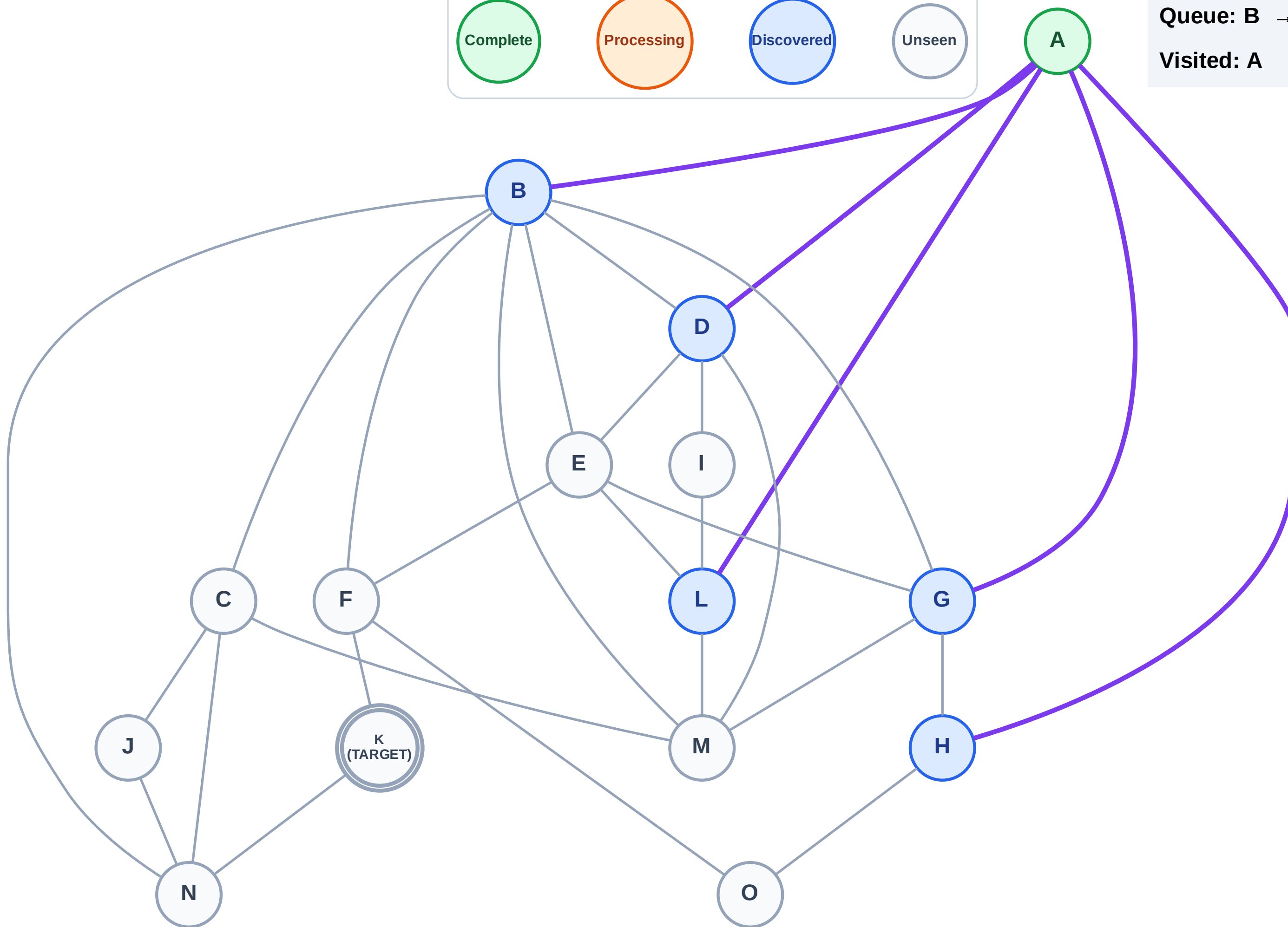
Step 3: Discover B, D, G, H, L from A

Legend



Queue: B → D → G → H → L

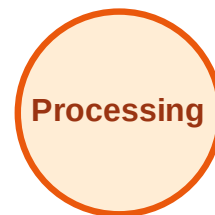
Visited: A



BFS Traversal

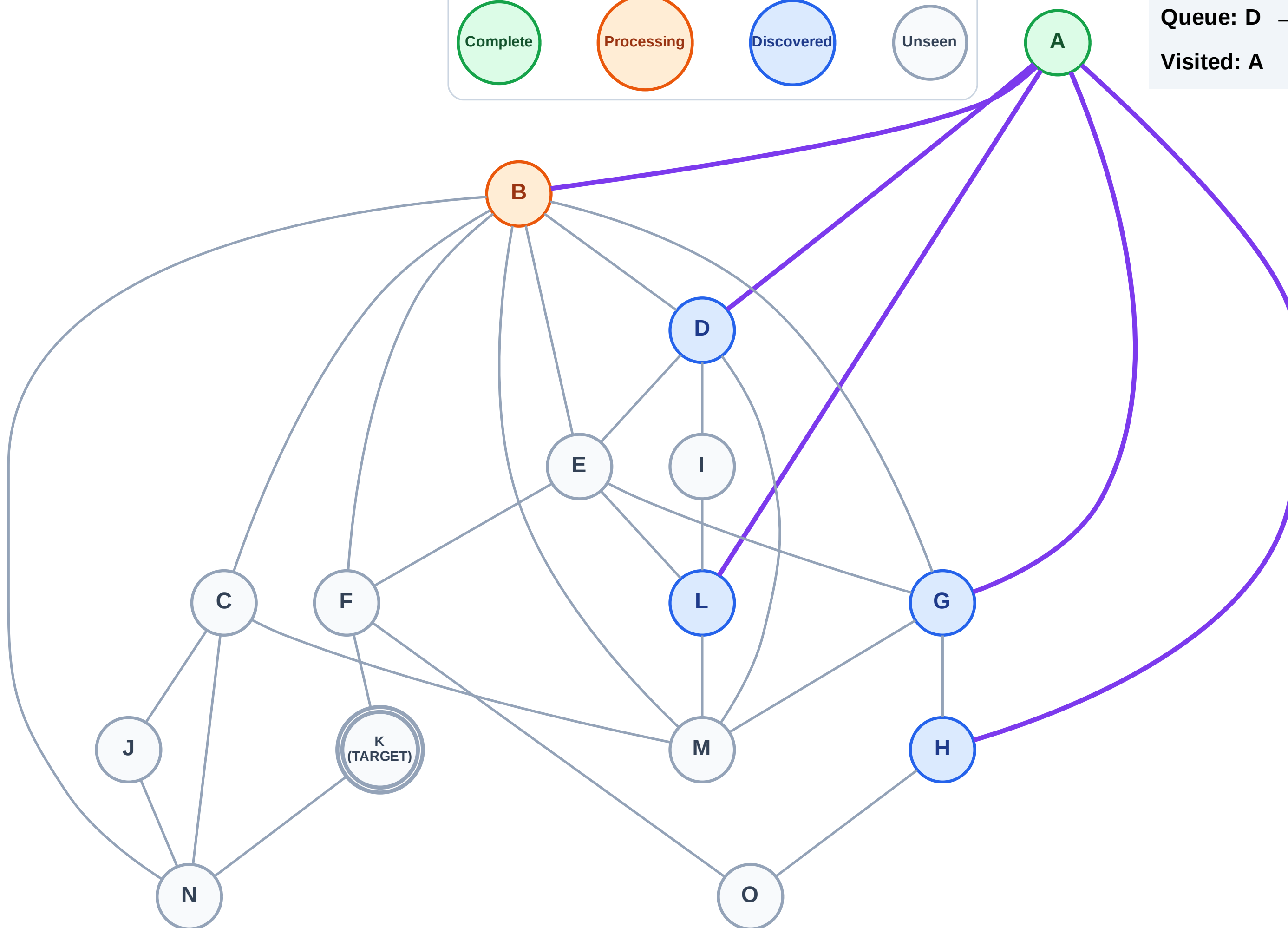
Step 4: Process node B

Legend



Queue: D → G → H → L

Visited: A



BFS Traversal

Step 5: Discover C, E, F, M, N from B

Legend



Complete



Processing



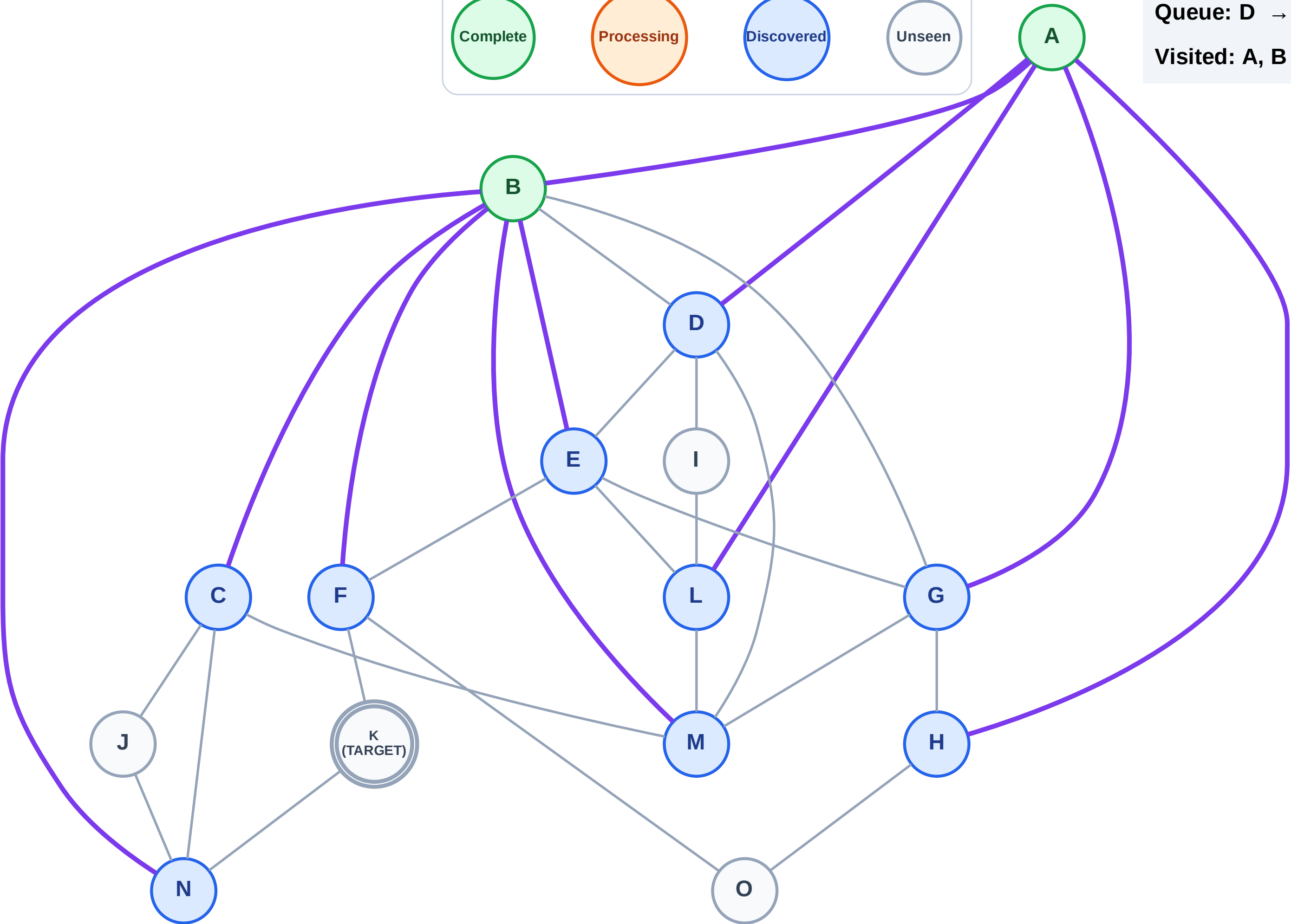
Discovered



Unseen

Queue: D → G → H → L → C → E → F → M → N

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

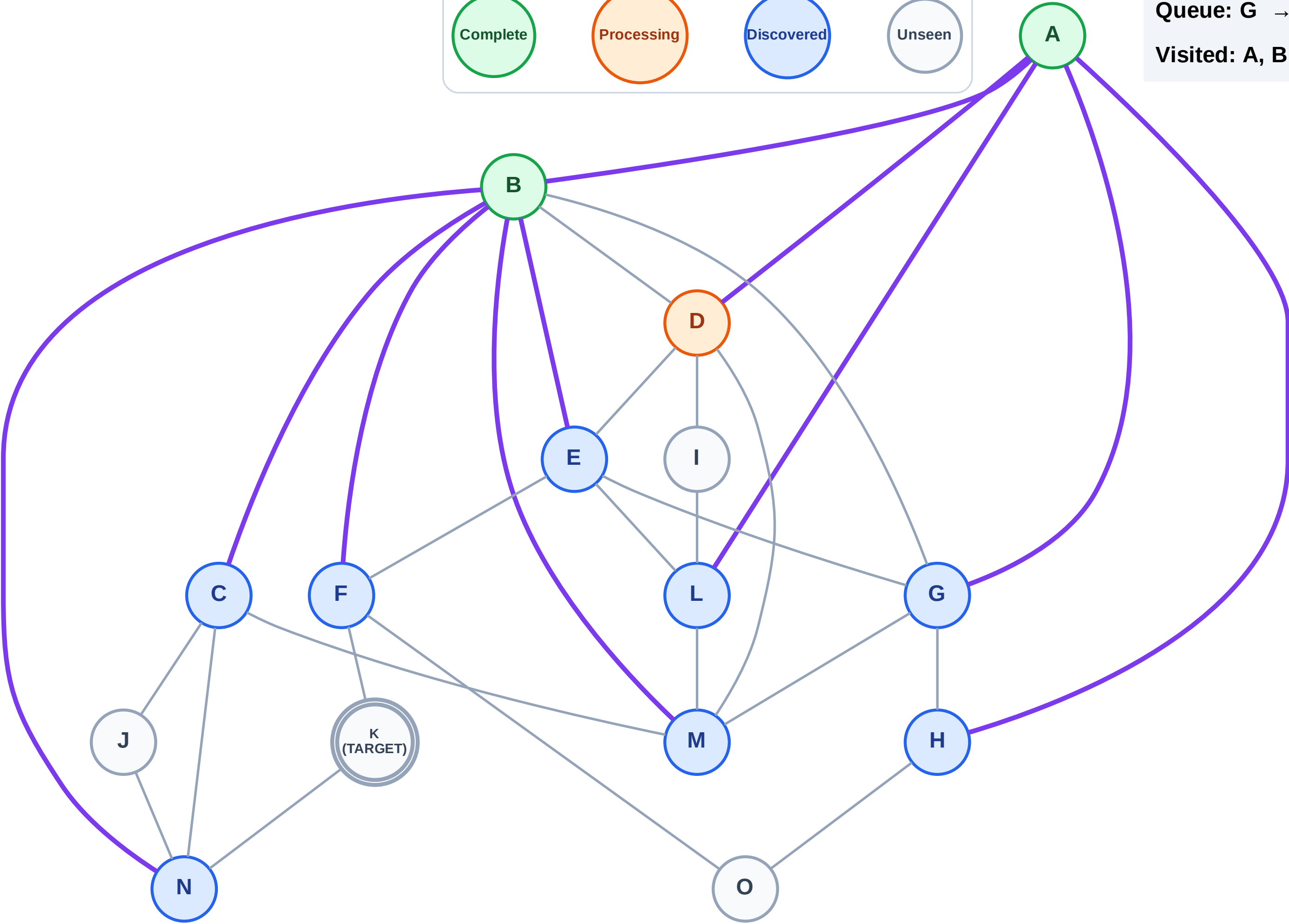
Processing

Discovered

Unseen

Queue: G → H → L → C → E → F → M → N

Visited: A, B



BFS Traversal

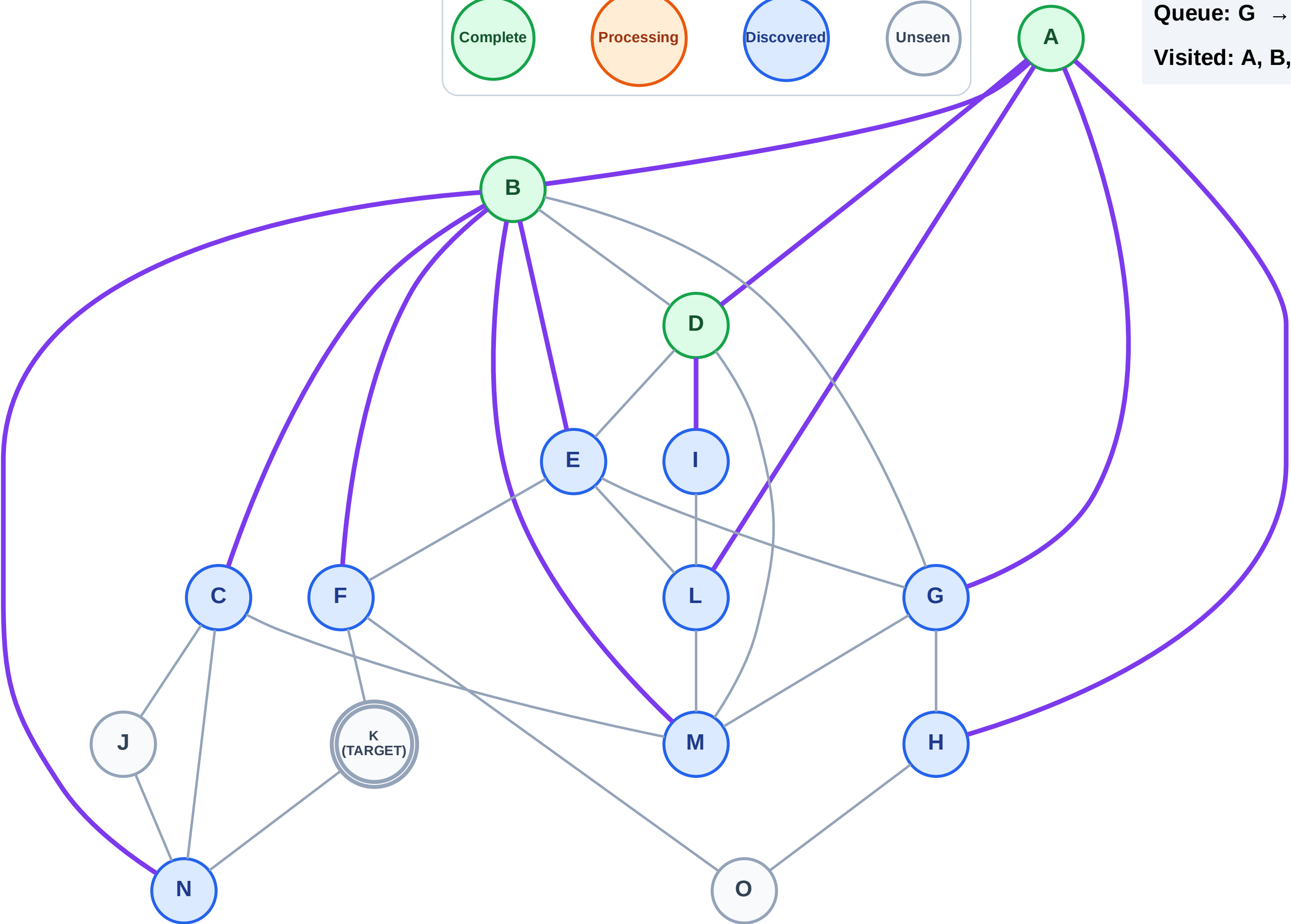
Step 7: Discover I from D

Legend



Queue: G → H → L → C → E → F → M → N → I

Visited: A, B, D



BFS Traversal

Step 8: Process node G

Legend

Complete

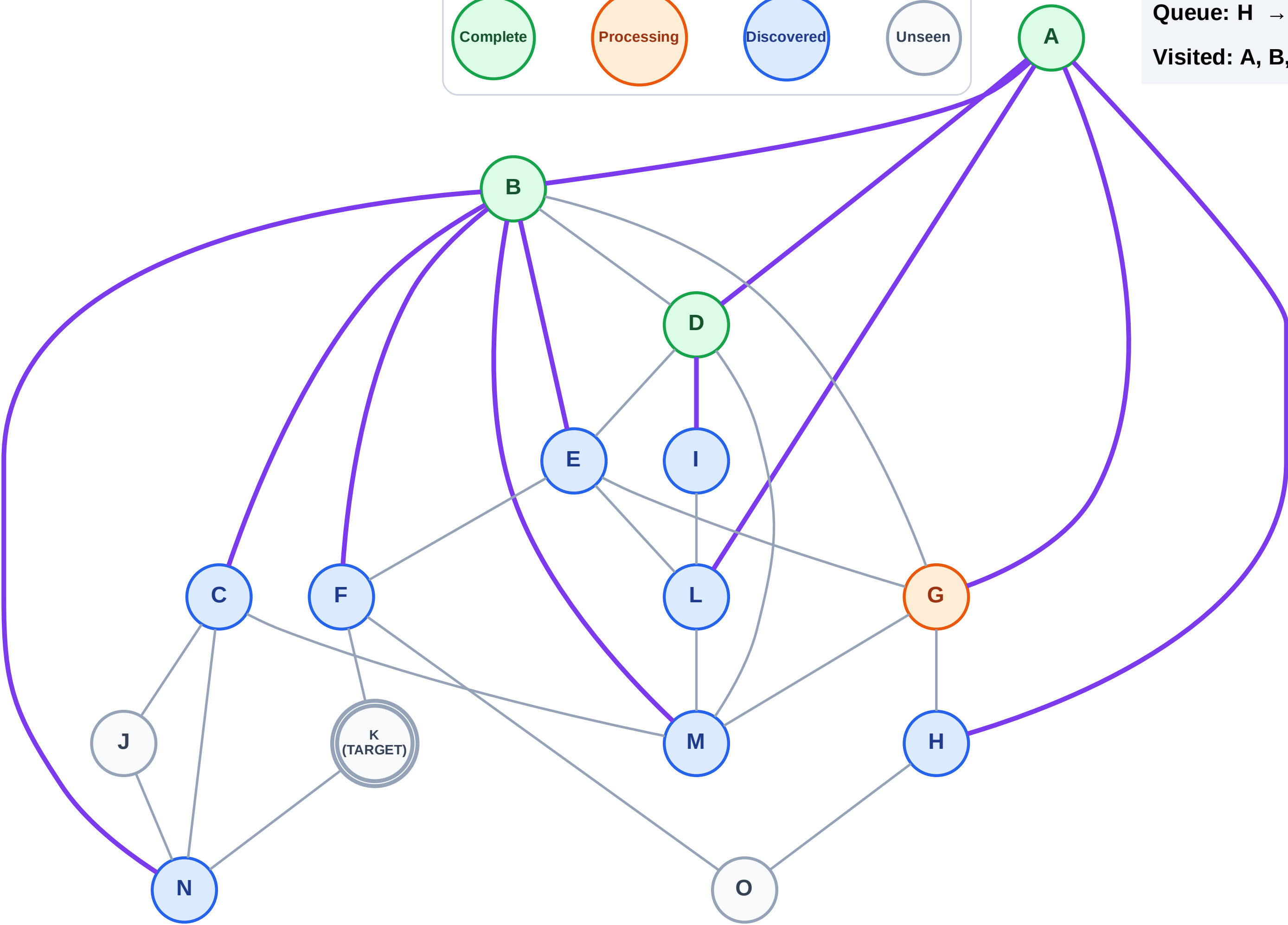
Processing

Discovered

Unseen

Queue: H → L → C → E → F → M → N → I

Visited: A, B, D



BFS Traversal

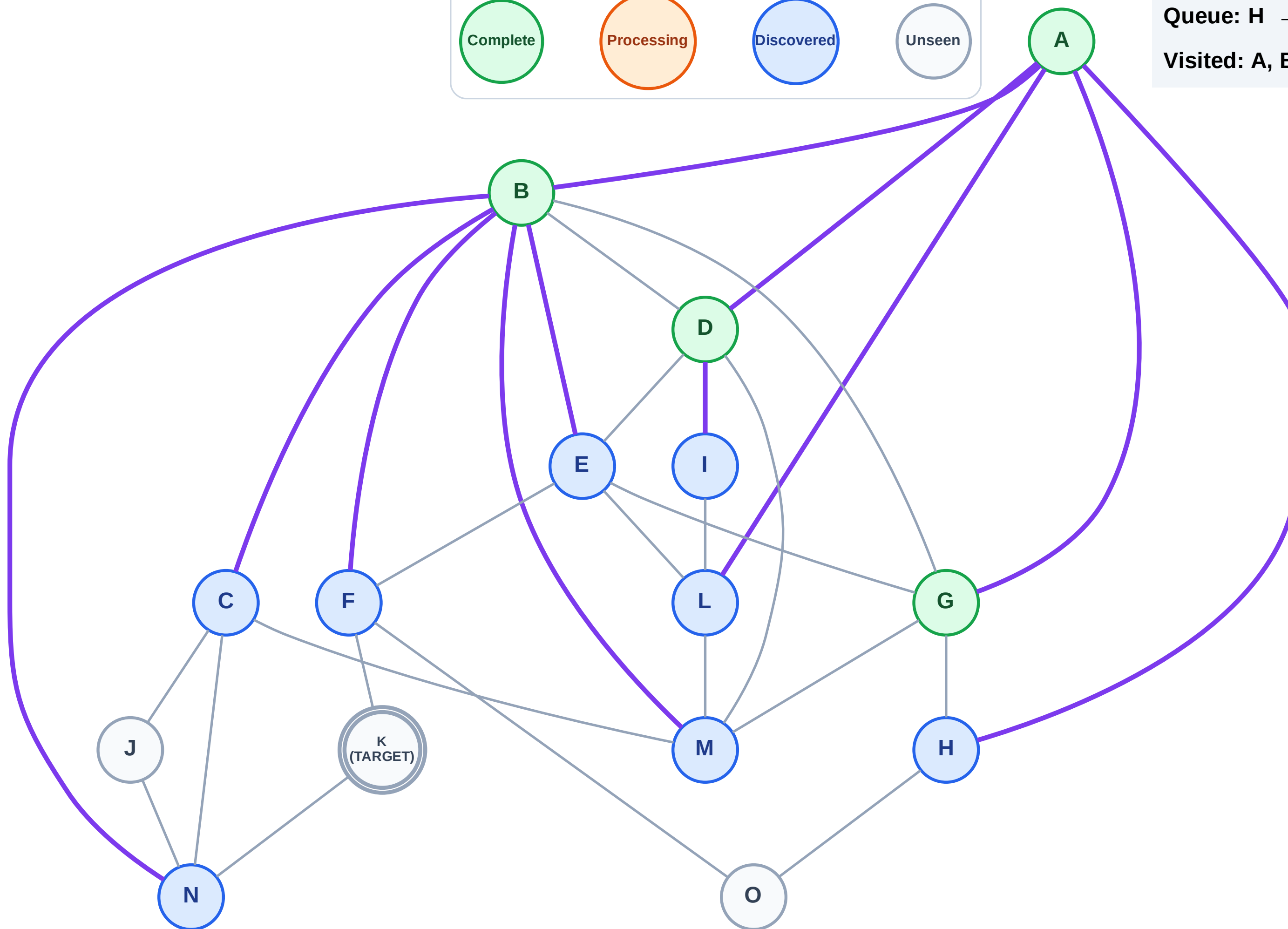
Step 9: No new neighbors from G

Legend



Queue: H → L → C → E → F → M → N → I

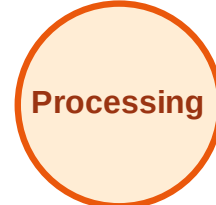
Visited: A, B, D, G



BFS Traversal

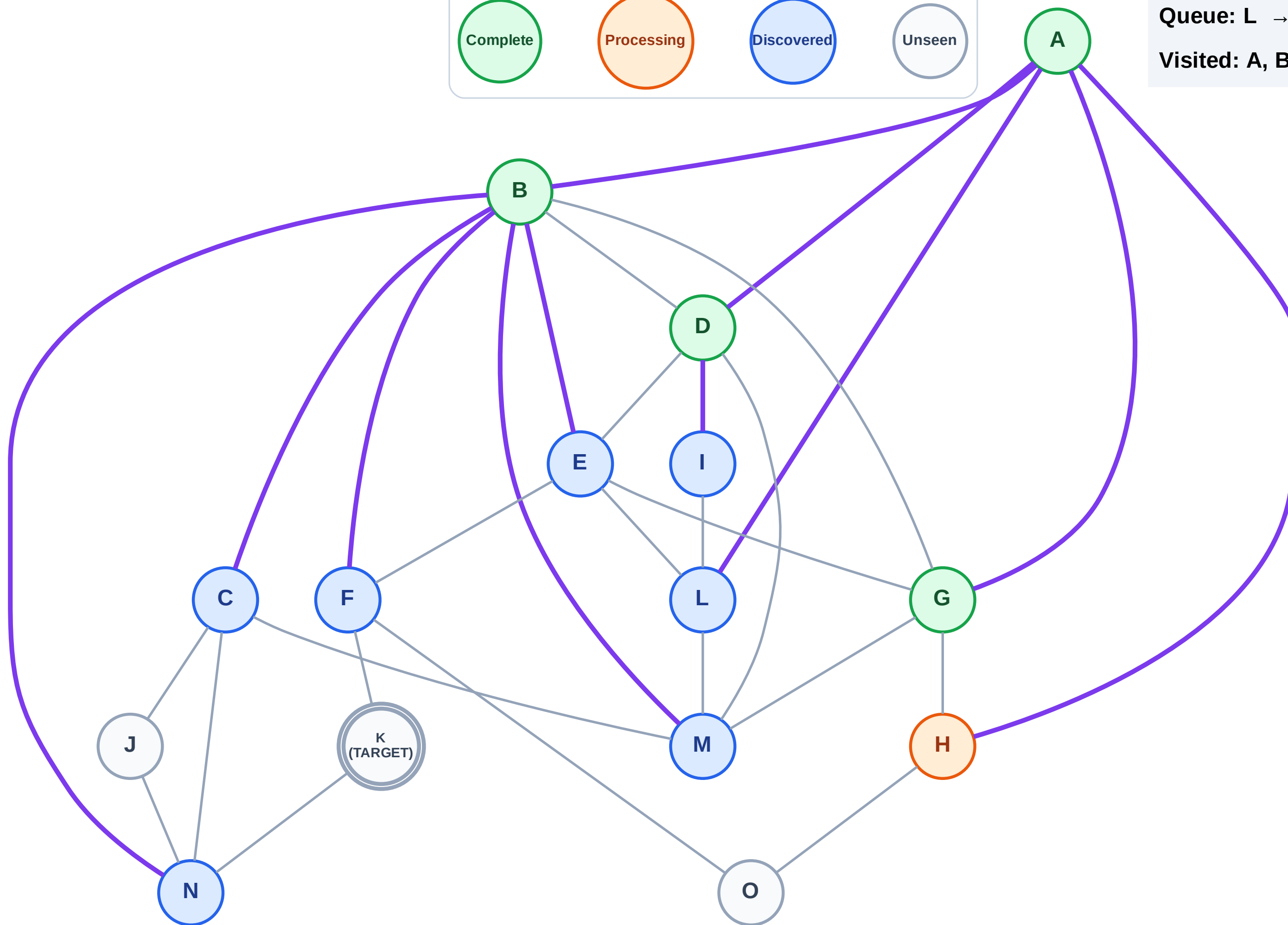
Step 10: Process node H

Legend



Queue: L → C → E → F → M → N → I

Visited: A, B, D, G



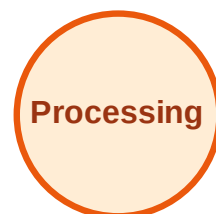
BFS Traversal

Step 11: Discover O from H

Legend



Complete



Processing



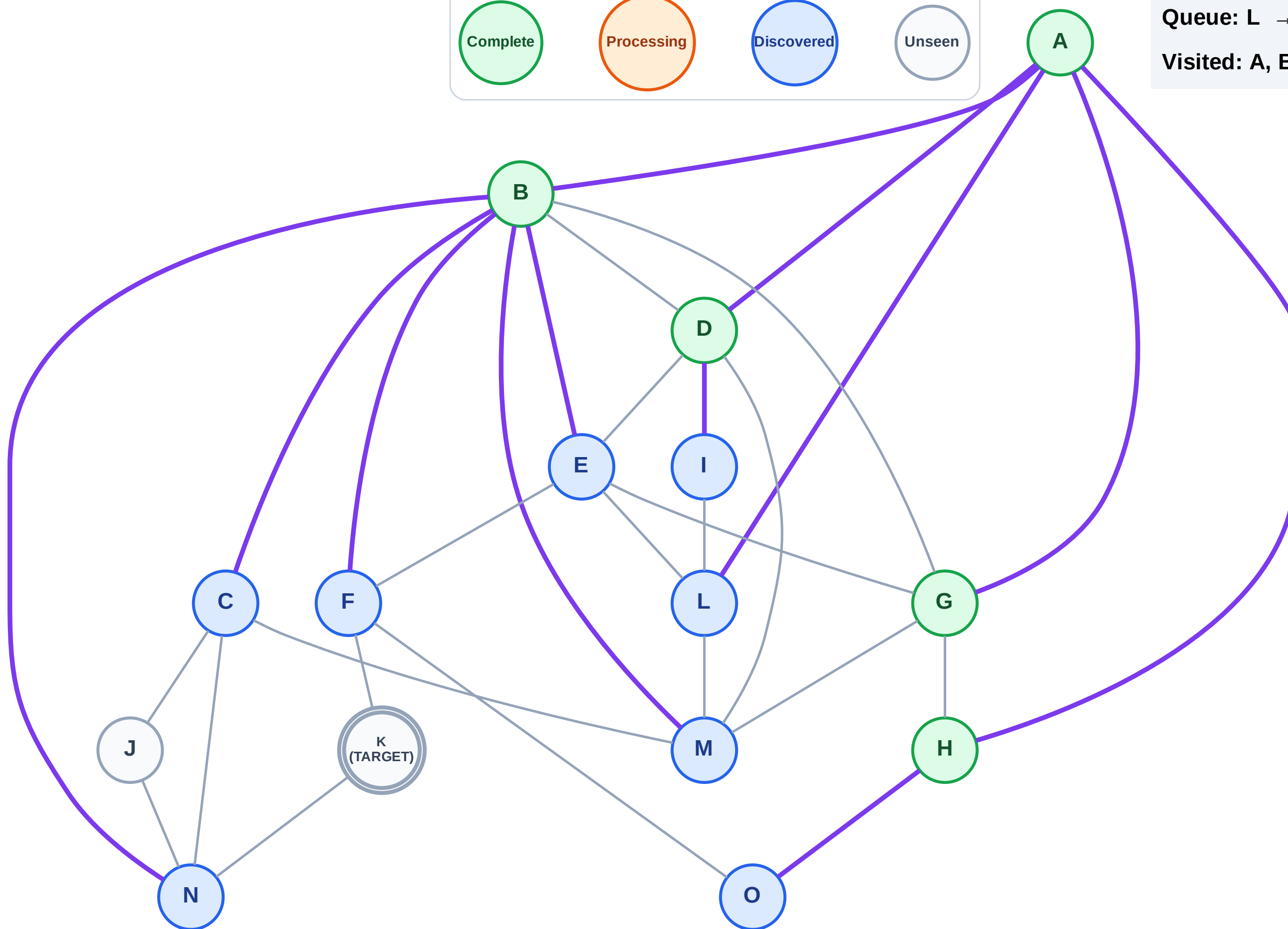
Discovered



Unseen

Queue: L → C → E → F → M → N → I → O

Visited: A, B, D, G, H



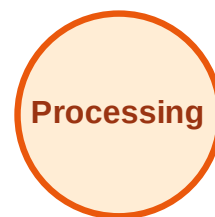
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing

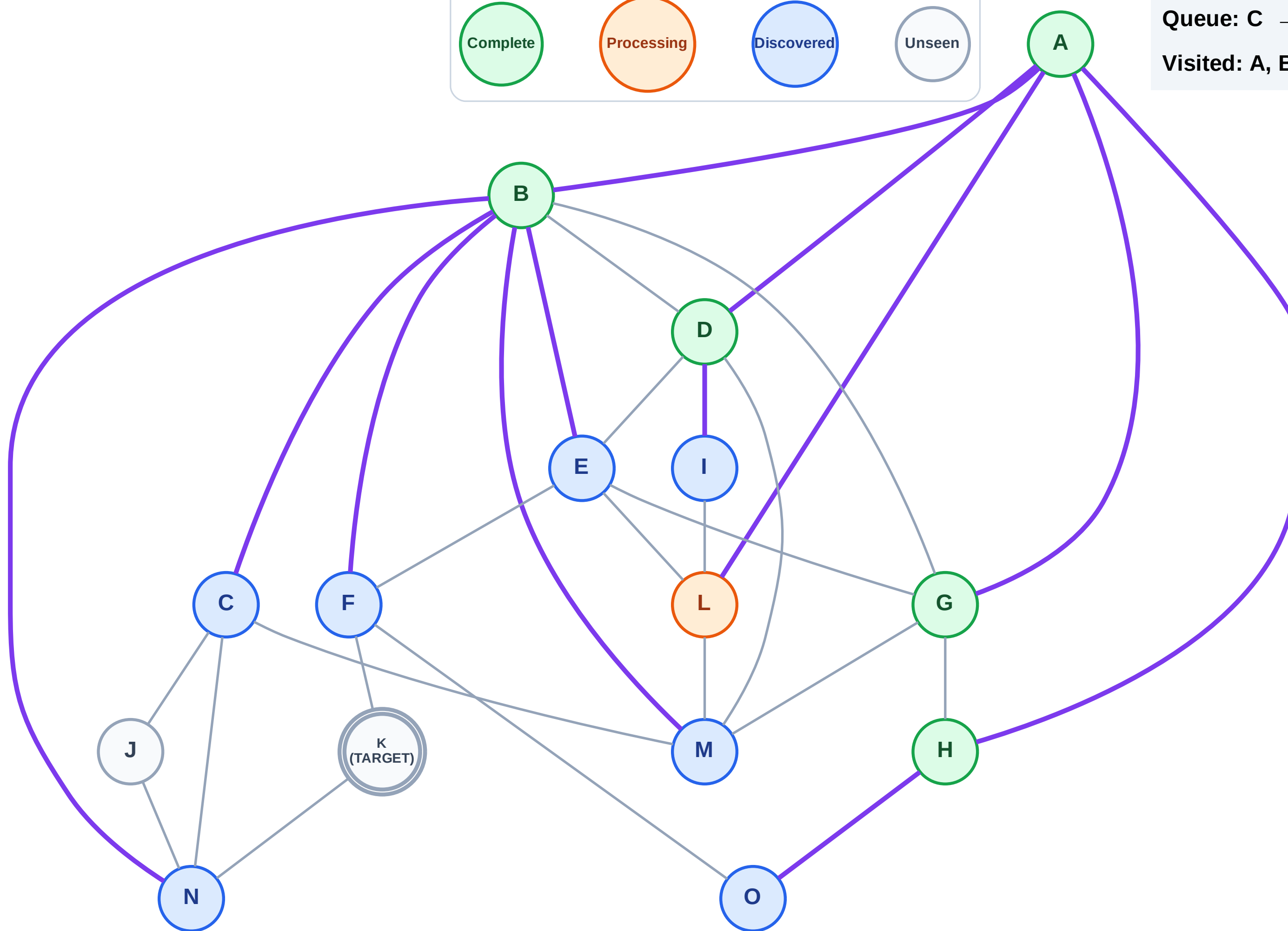


Discovered



Unseen

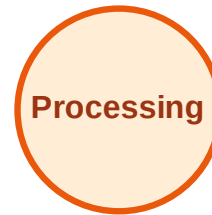
Queue: C → E → F → M → N → I → O
Visited: A, B, D, G, H



BFS Traversal

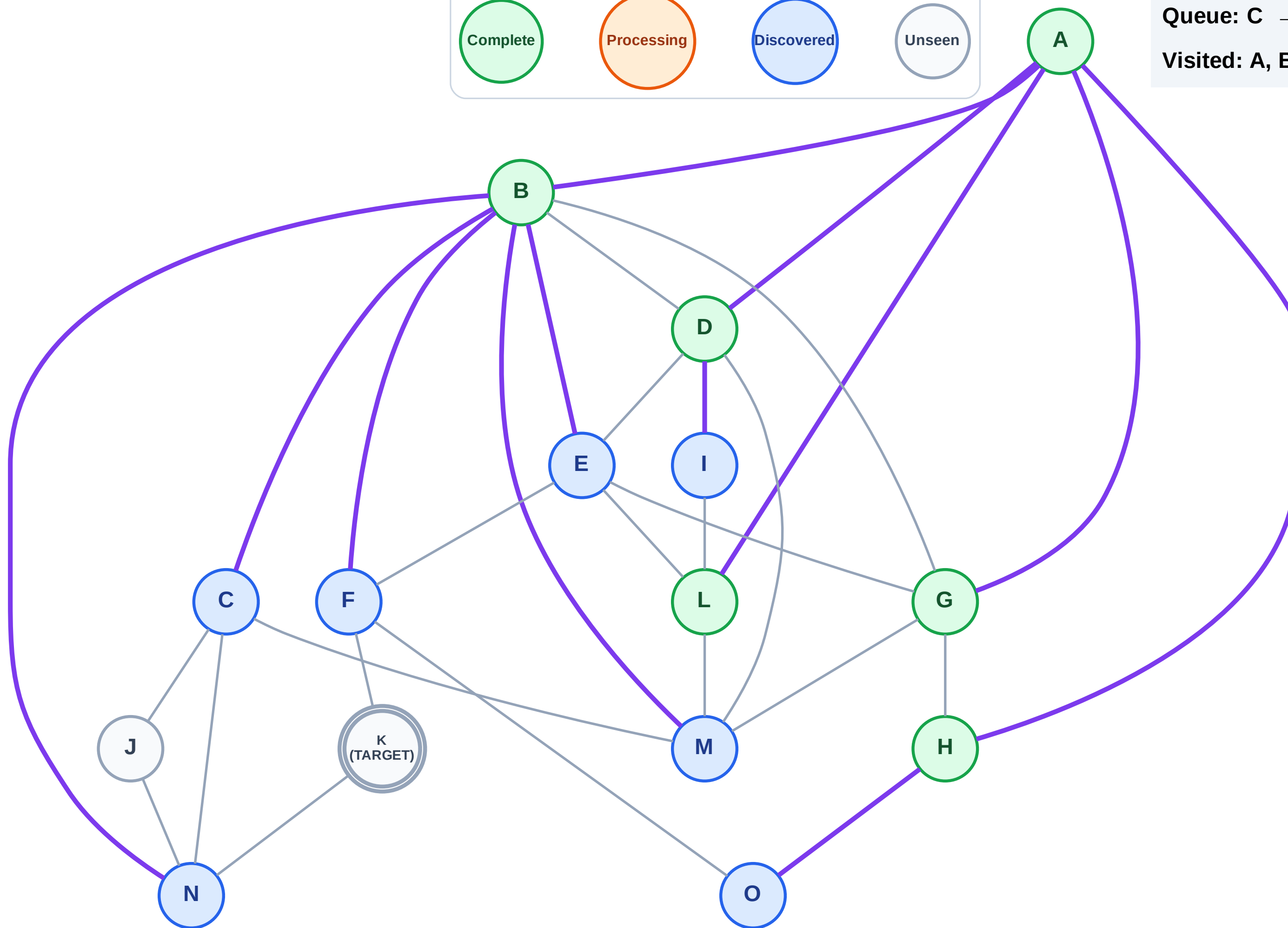
Step 13: No new neighbors from L

Legend



Queue: C → E → F → M → N → I → O

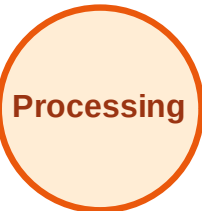
Visited: A, B, D, G, H, L



BFS Traversal

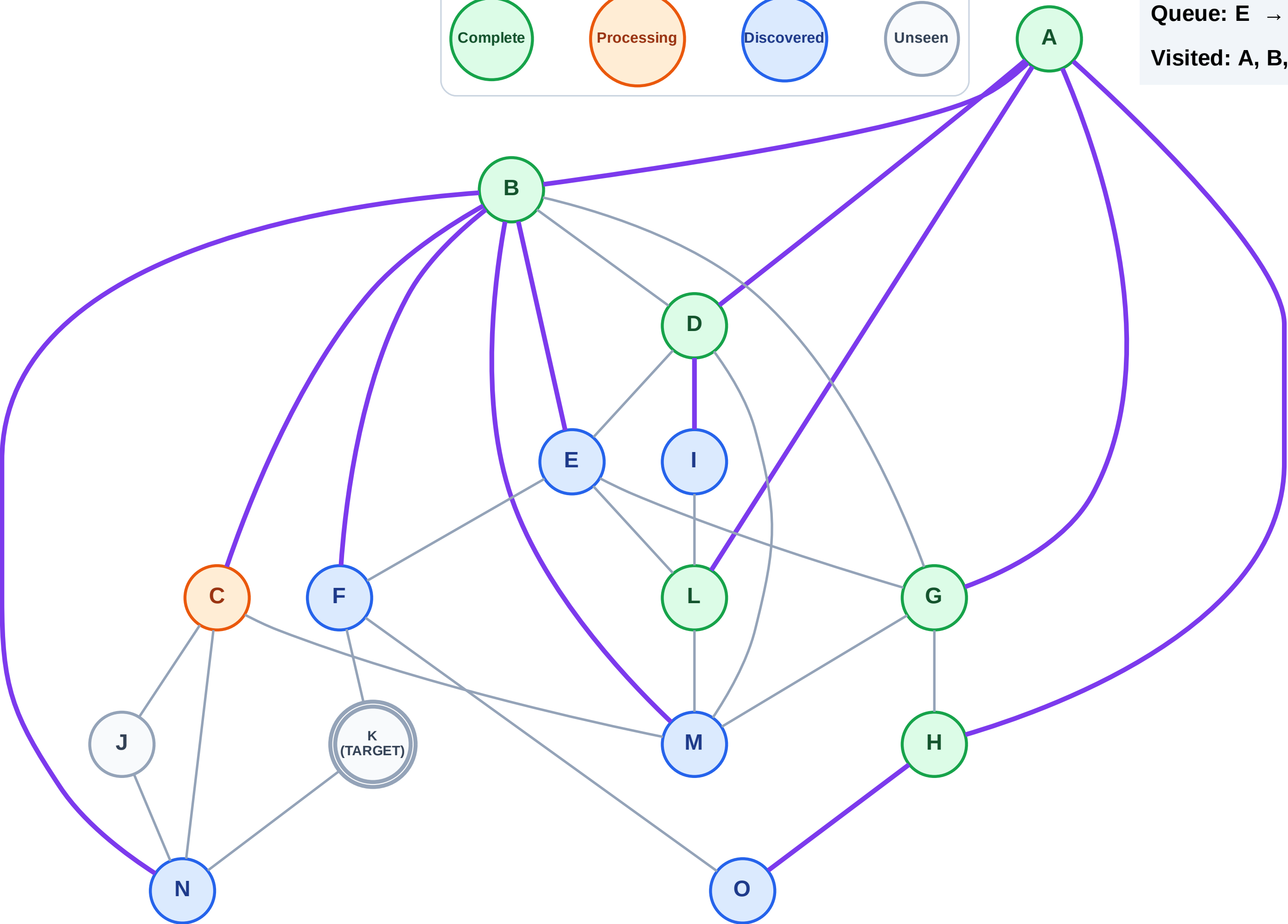
Step 14: Process node C

Legend



Queue: E → F → M → N → I → O

Visited: A, B, D, G, H, L



BFS Traversal

Step 15: Discover J from C

Legend

Complete

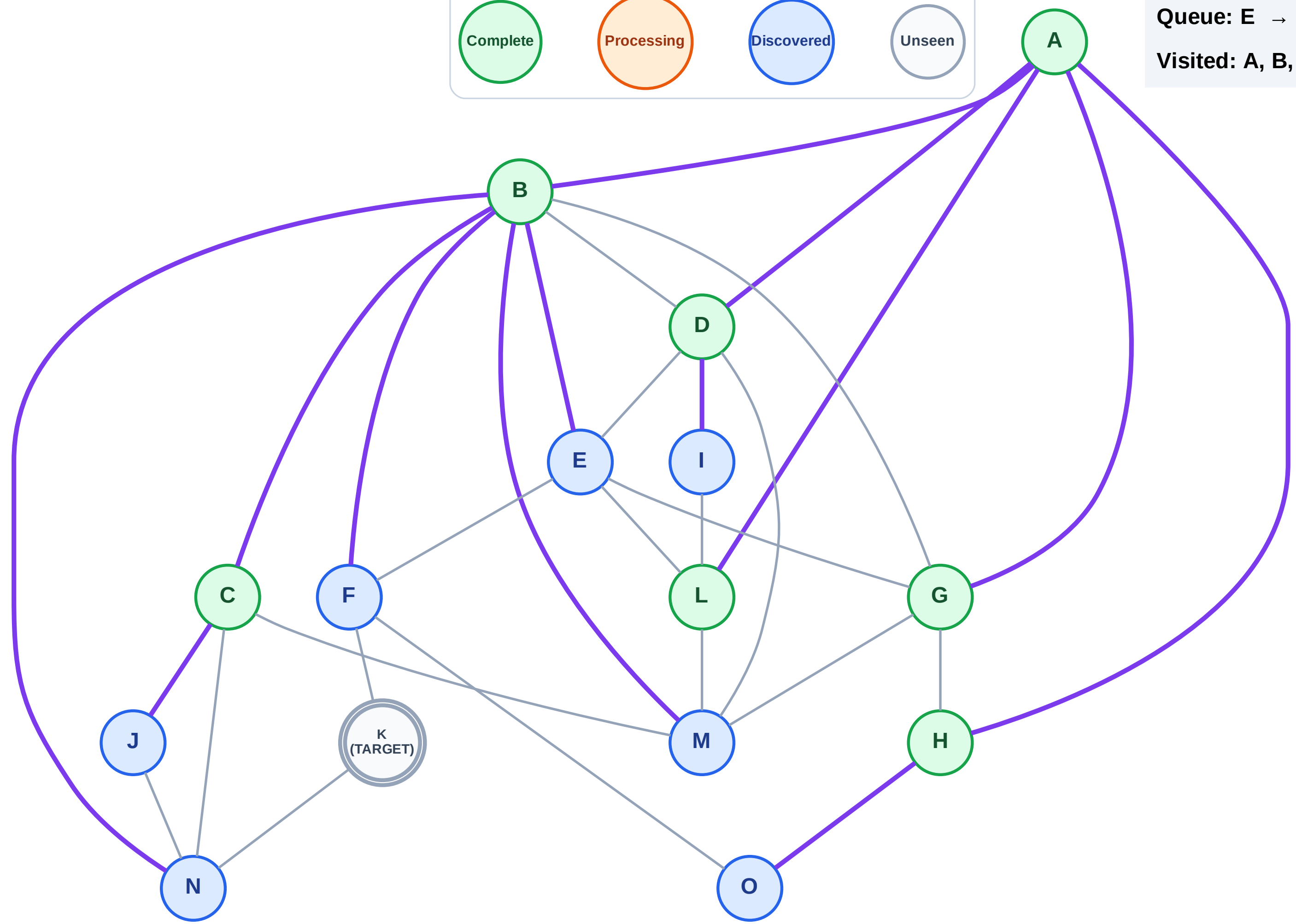
Processing

Discovered

Unseen

Queue: E → F → M → N → I → O → J

Visited: A, B, C, D, G, H, L



BFS Traversal

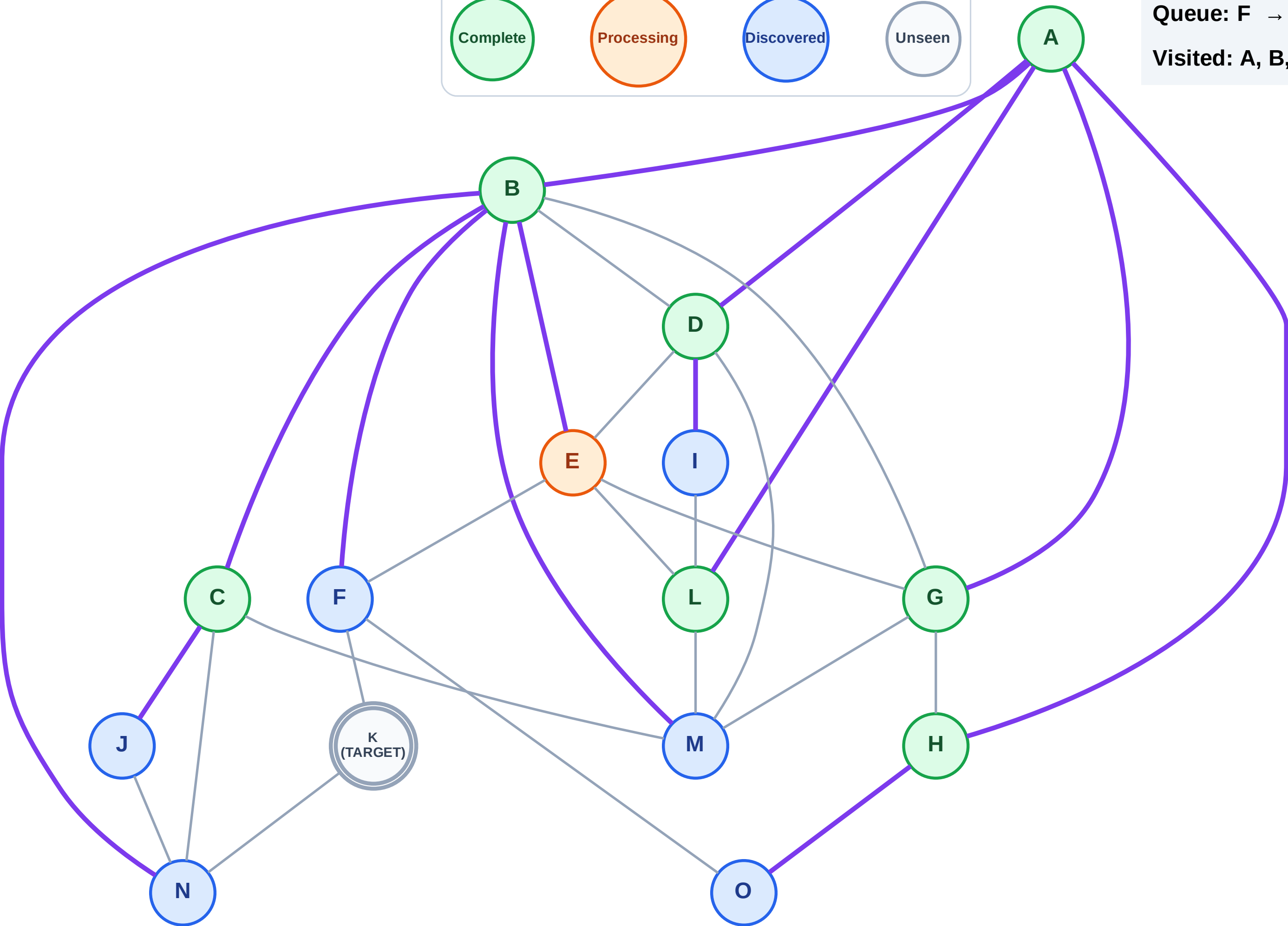
Step 16: Process node E

Legend



Queue: F → M → N → I → O → J

Visited: A, B, C, D, G, H, L



BFS Traversal

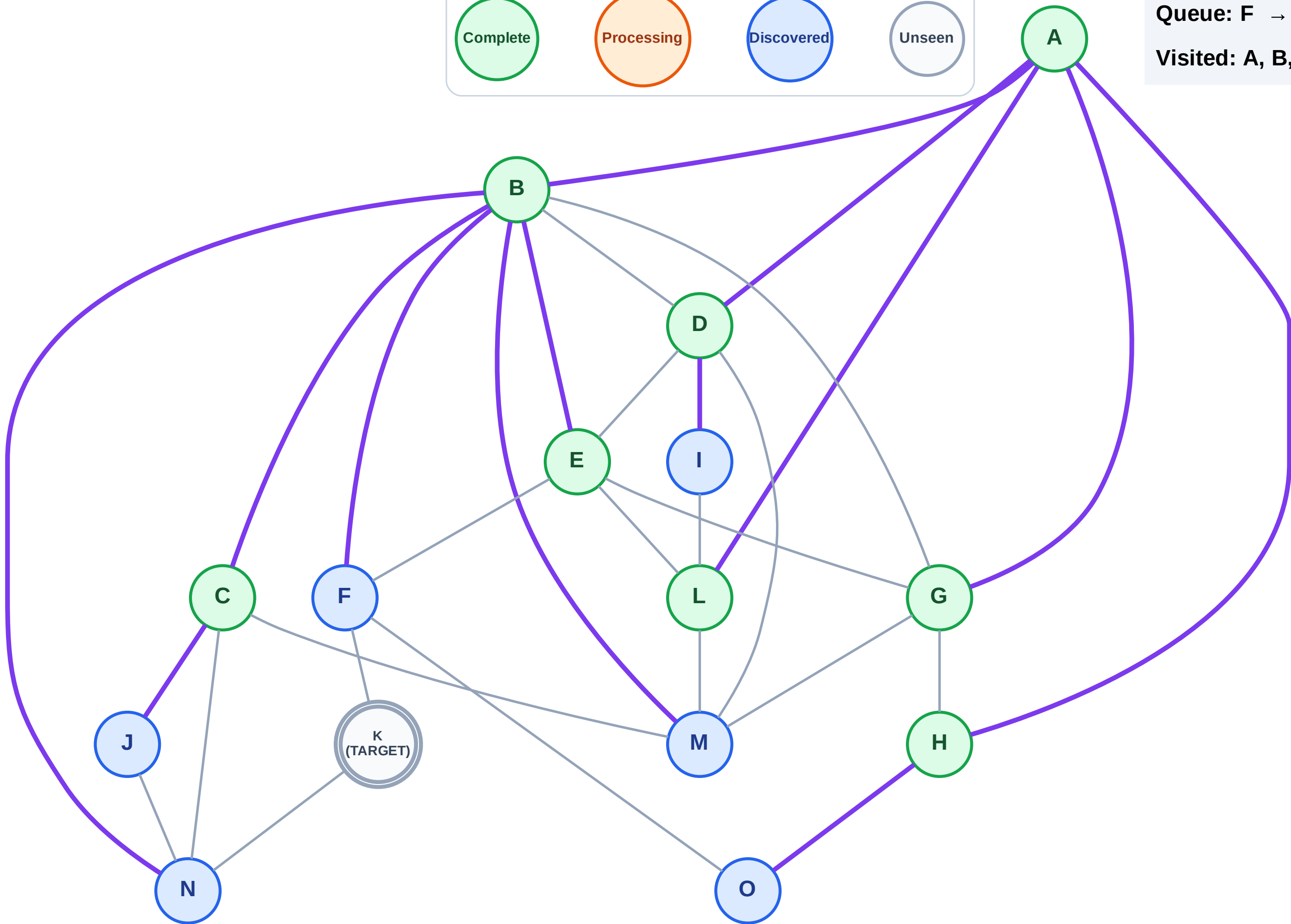
Step 17: No new neighbors from E

Legend



Queue: F → M → N → I → O → J

Visited: A, B, C, D, E, G, H, L



BFS Traversal

Step 18: Process node F

Legend

Complete

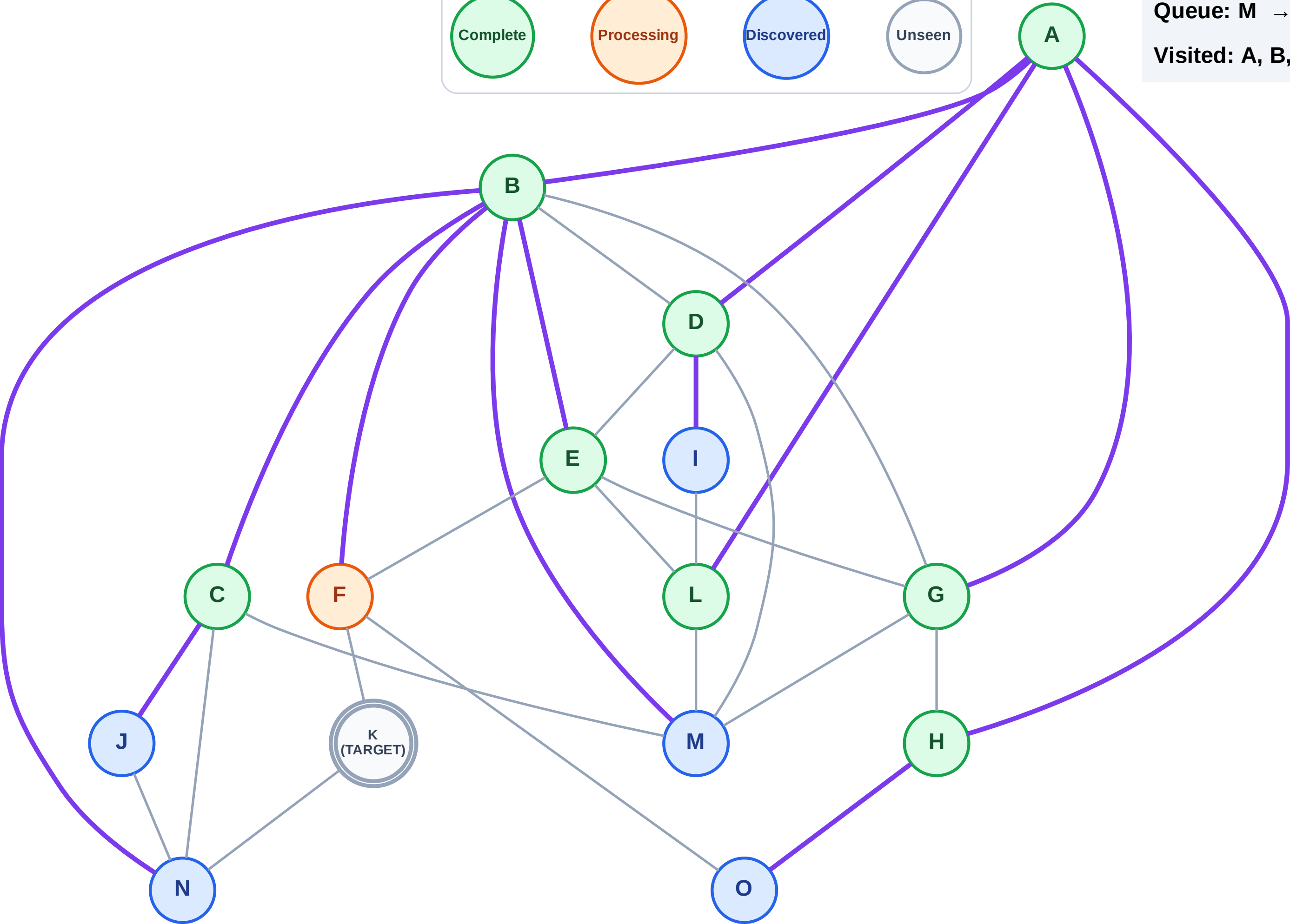
Processing

Discovered

Unseen

Queue: M → N → I → O → J

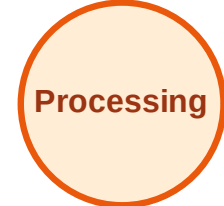
Visited: A, B, C, D, E, G, H, L



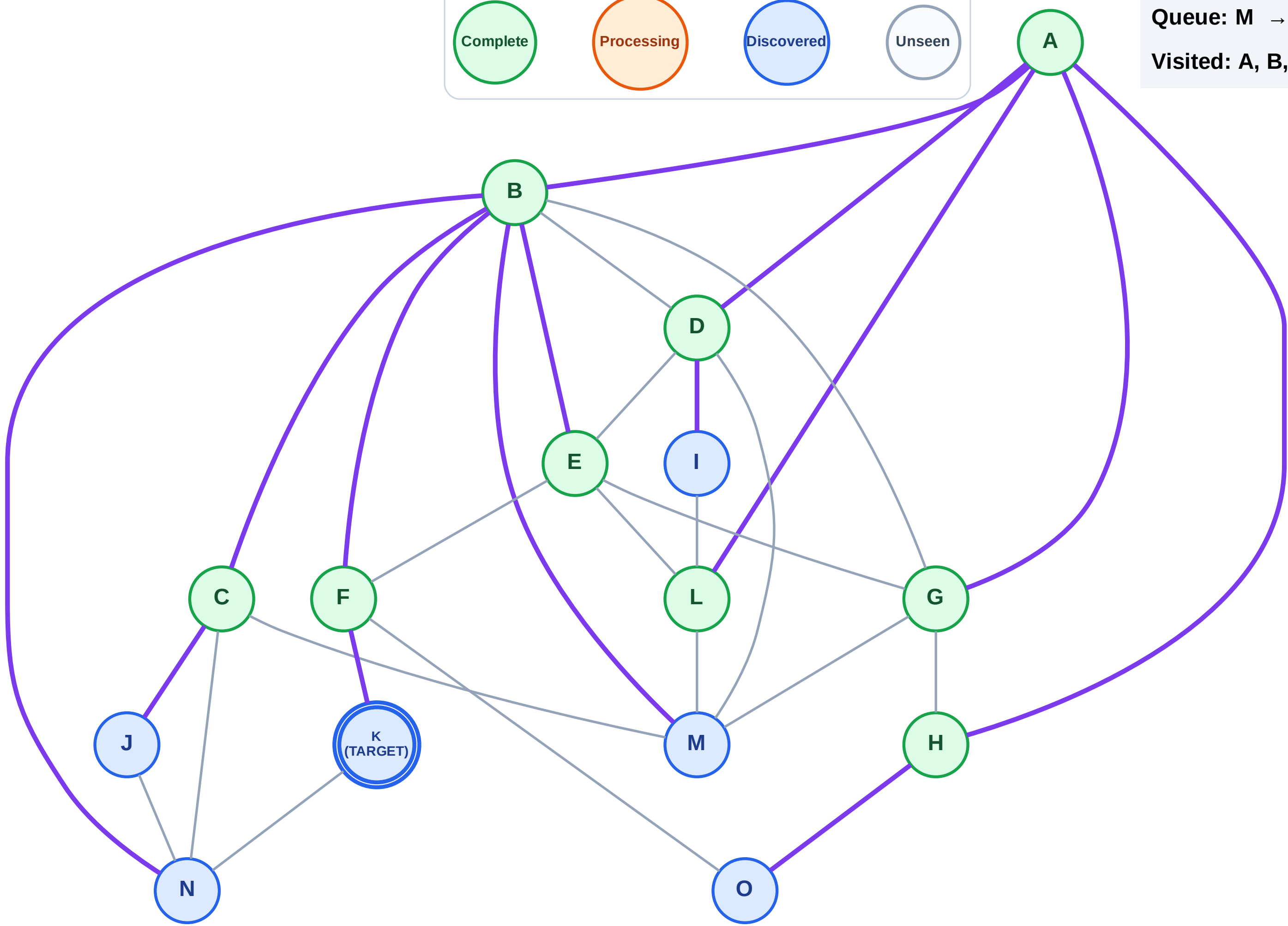
BFS Traversal

Step 19: Discover K from F

Legend



Queue: M → N → I → O → J → K
Visited: A, B, C, D, E, F, G, H, L



BFS Traversal

Step 20: Process node M

Legend

Complete

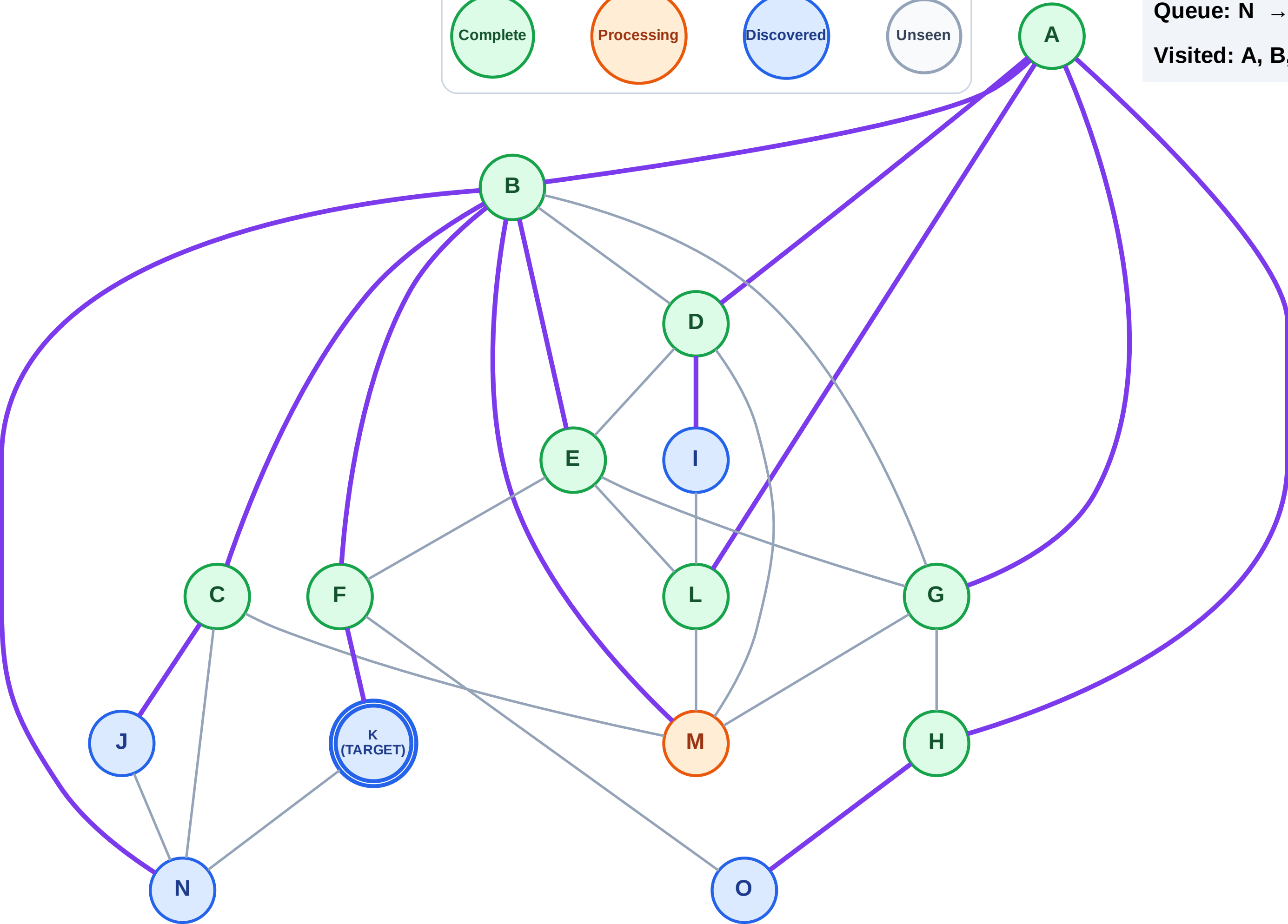
Processing

Discovered

Unseen

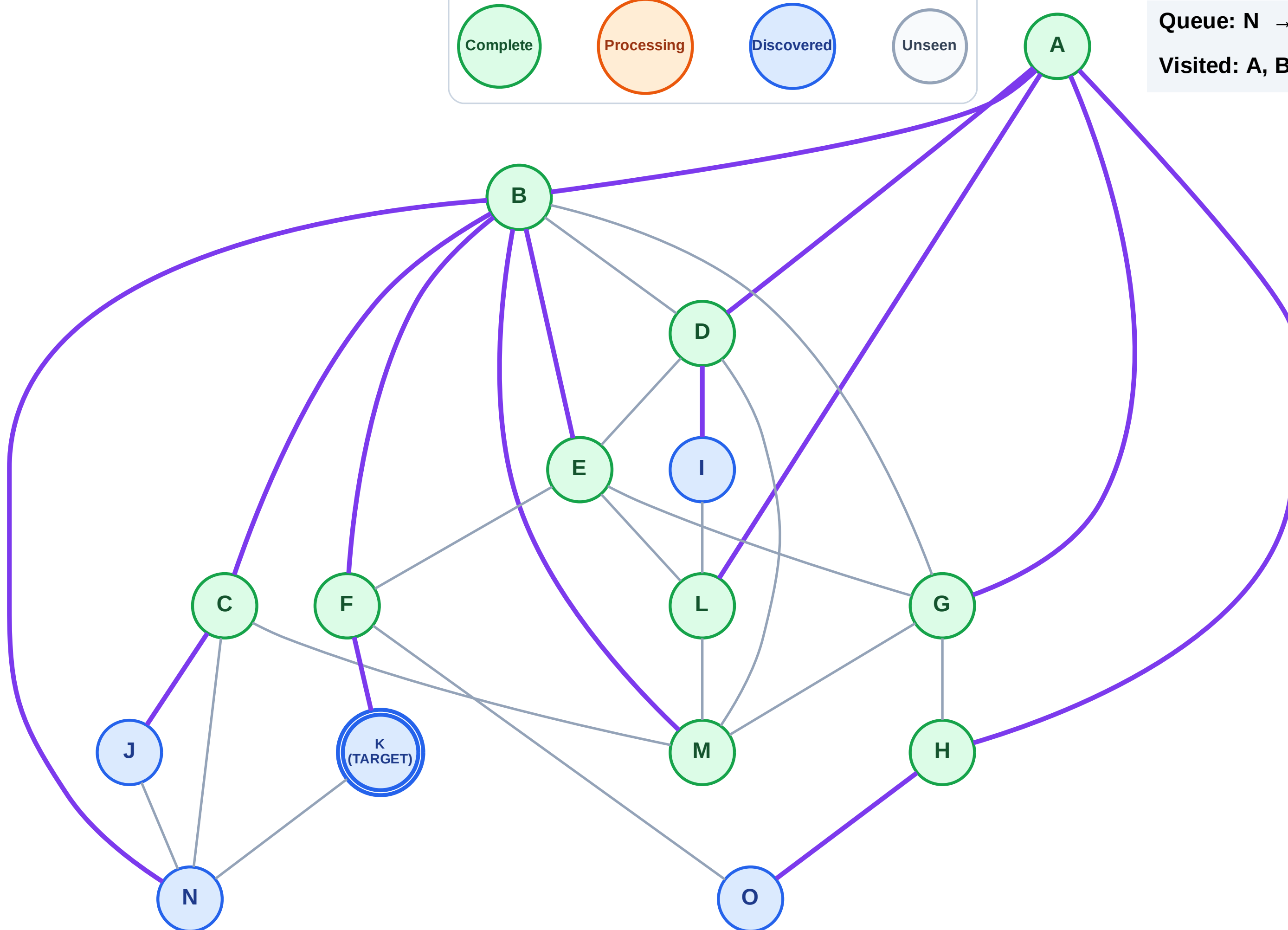
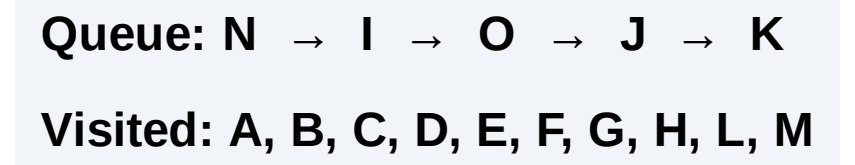
Queue: N → I → O → J → K

Visited: A, B, C, D, E, F, G, H, L



Step 21: No new neighbors from M

Legend



BFS Traversal

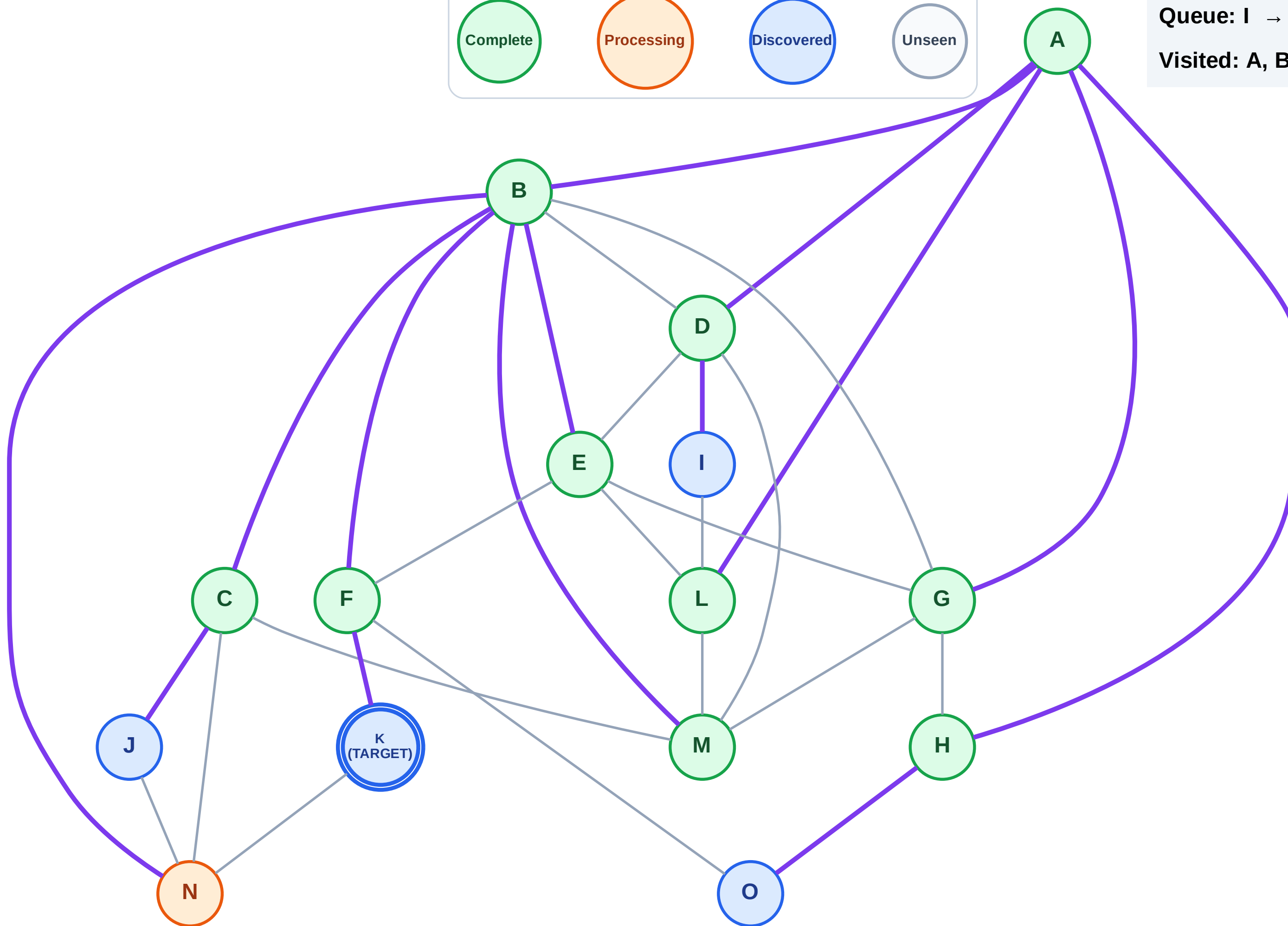
Step 22: Process node N

Legend



Queue: I → O → J → K

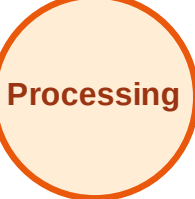
Visited: A, B, C, D, E, F, G, H, L, M



BFS Traversal

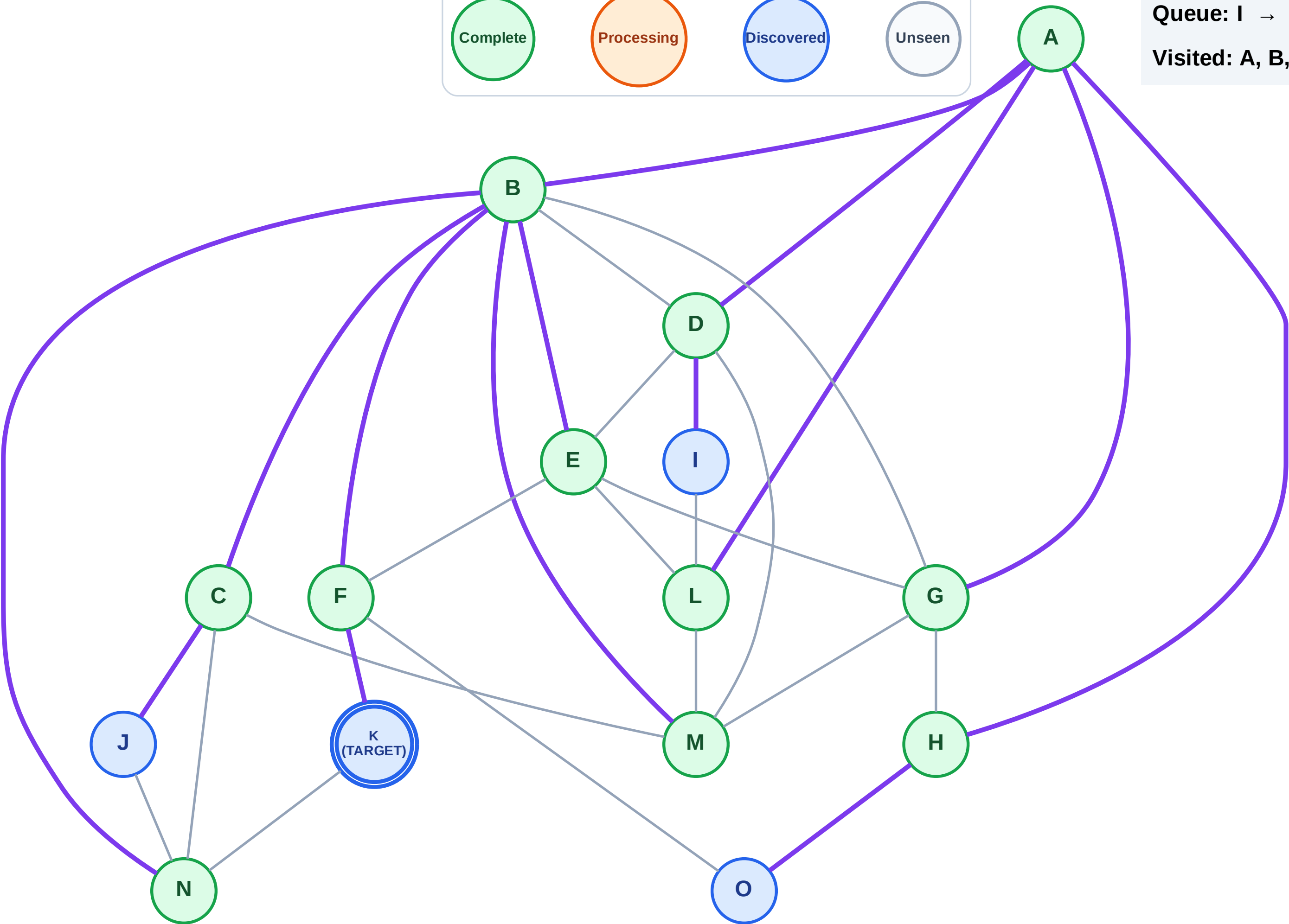
Step 23: No new neighbors from N

Legend



Queue: I → O → J → K

Visited: A, B, C, D, E, F, G, H, L, M, N



BFS Traversal

Step 24: Process node I

Legend

Complete

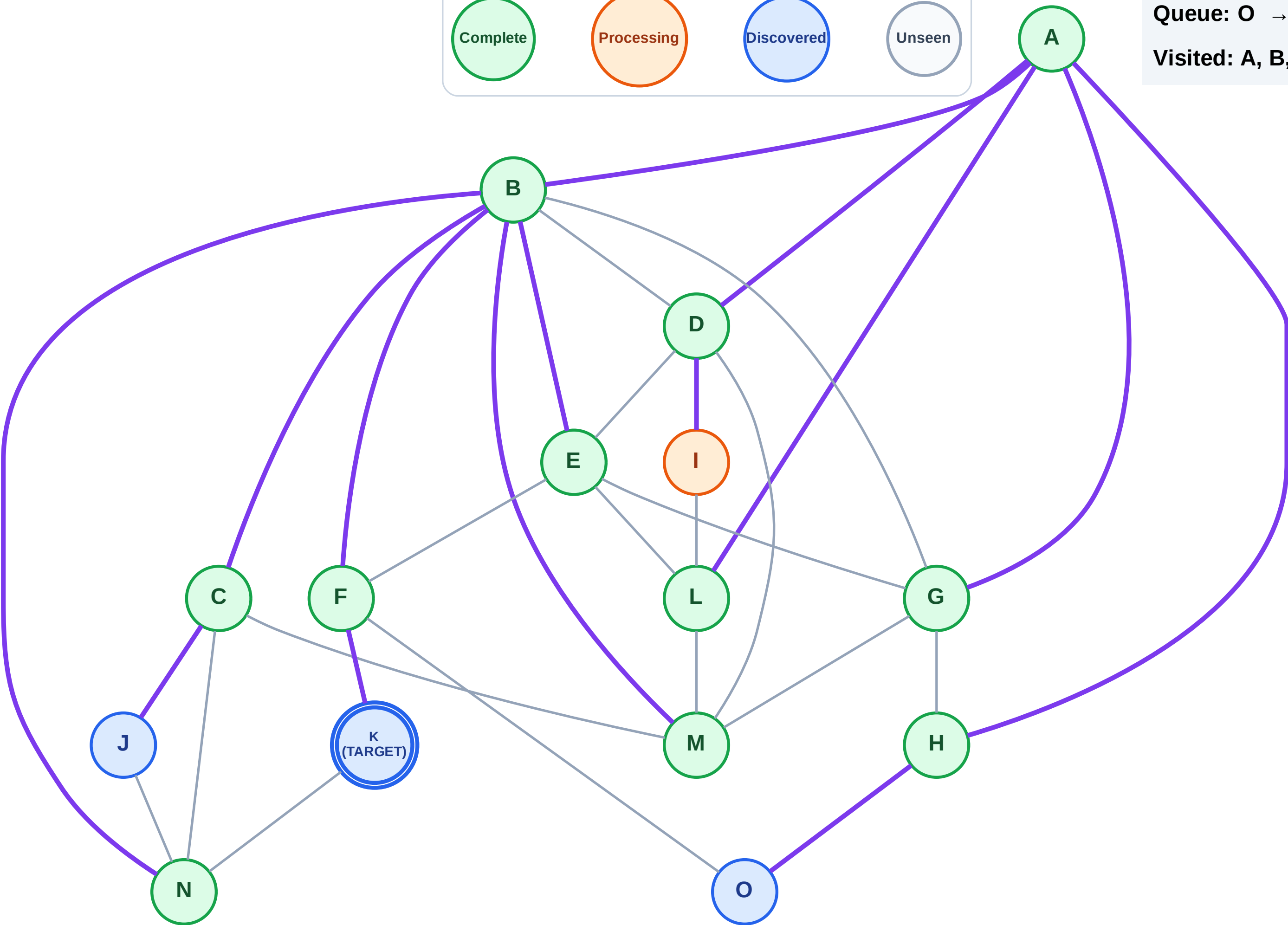
Processing

Discovered

Unseen

Queue: O → J → K

Visited: A, B, C, D, E, F, G, H, L, M, N



BFS Traversal

Step 25: No new neighbors from I

Legend



Complete



Processing



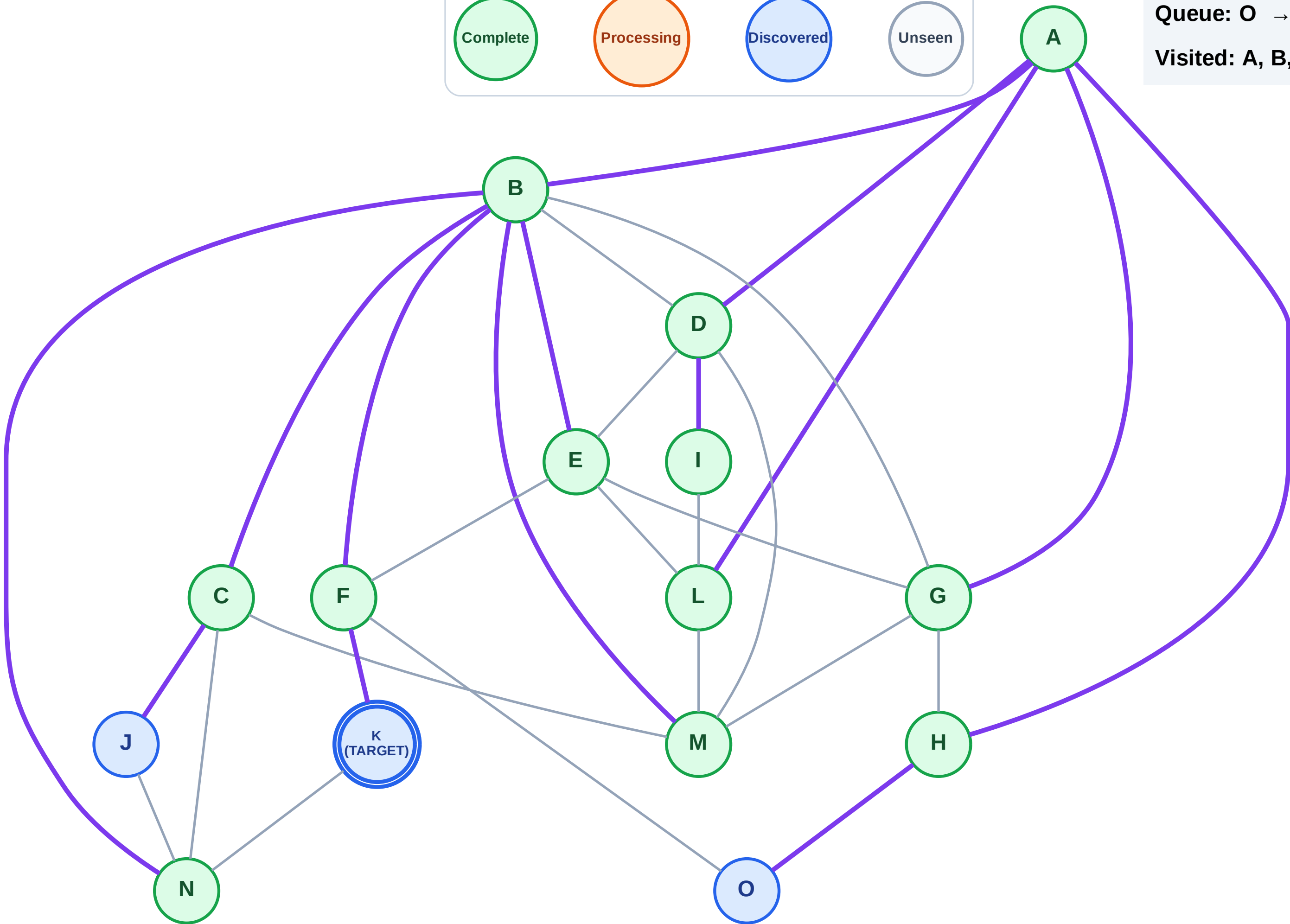
Discovered



Unseen

Queue: O → J → K

Visited: A, B, C, D, E, F, G, H, I, L, M, N



BFS Traversal

Step 26: Process node O

Legend



Complete



Processing



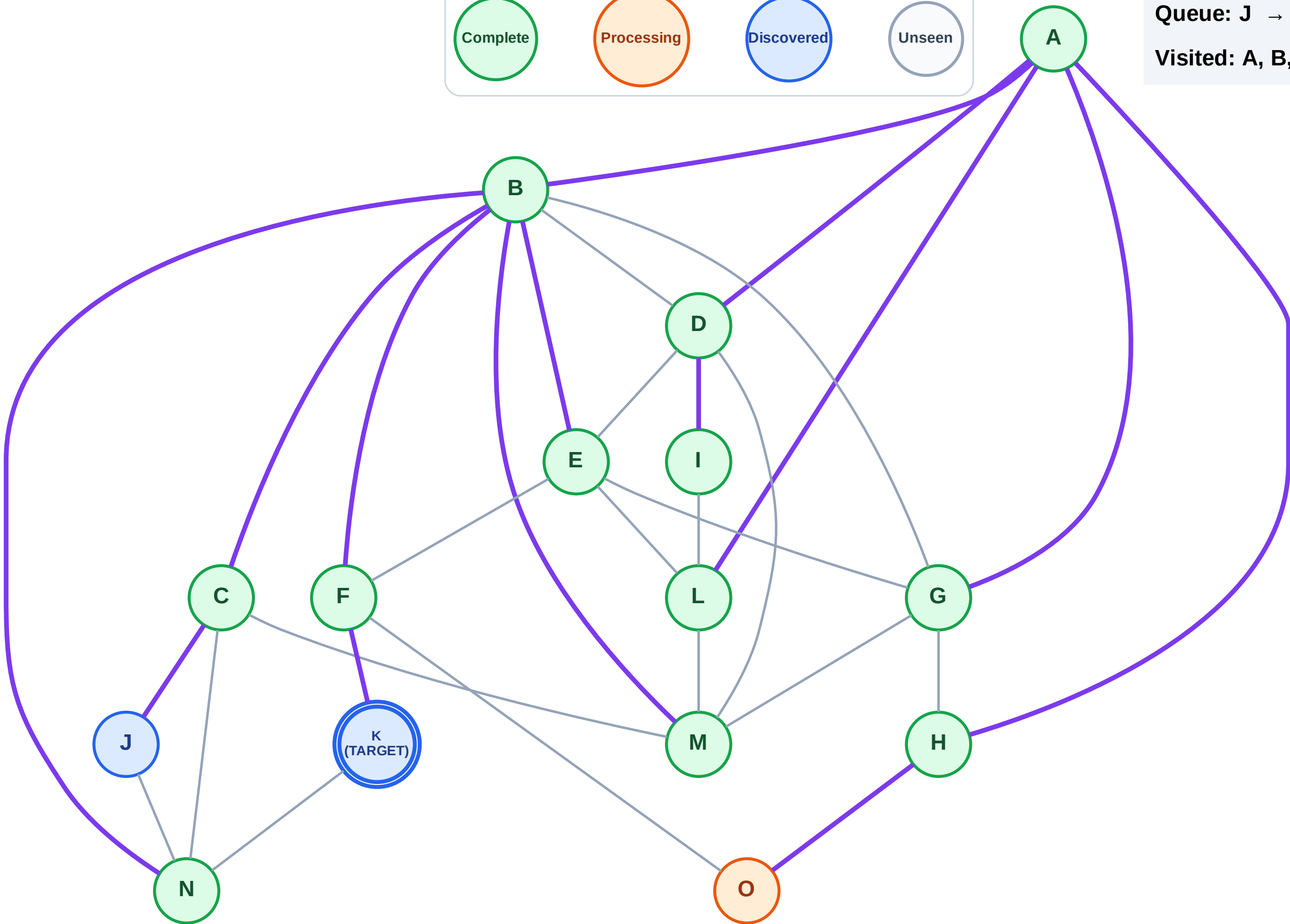
Discovered



Unseen

Queue: J → K

Visited: A, B, C, D, E, F, G, H, I, L, M, N



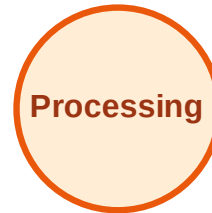
BFS Traversal

Step 27: No new neighbors from O

Legend



Complete



Processing



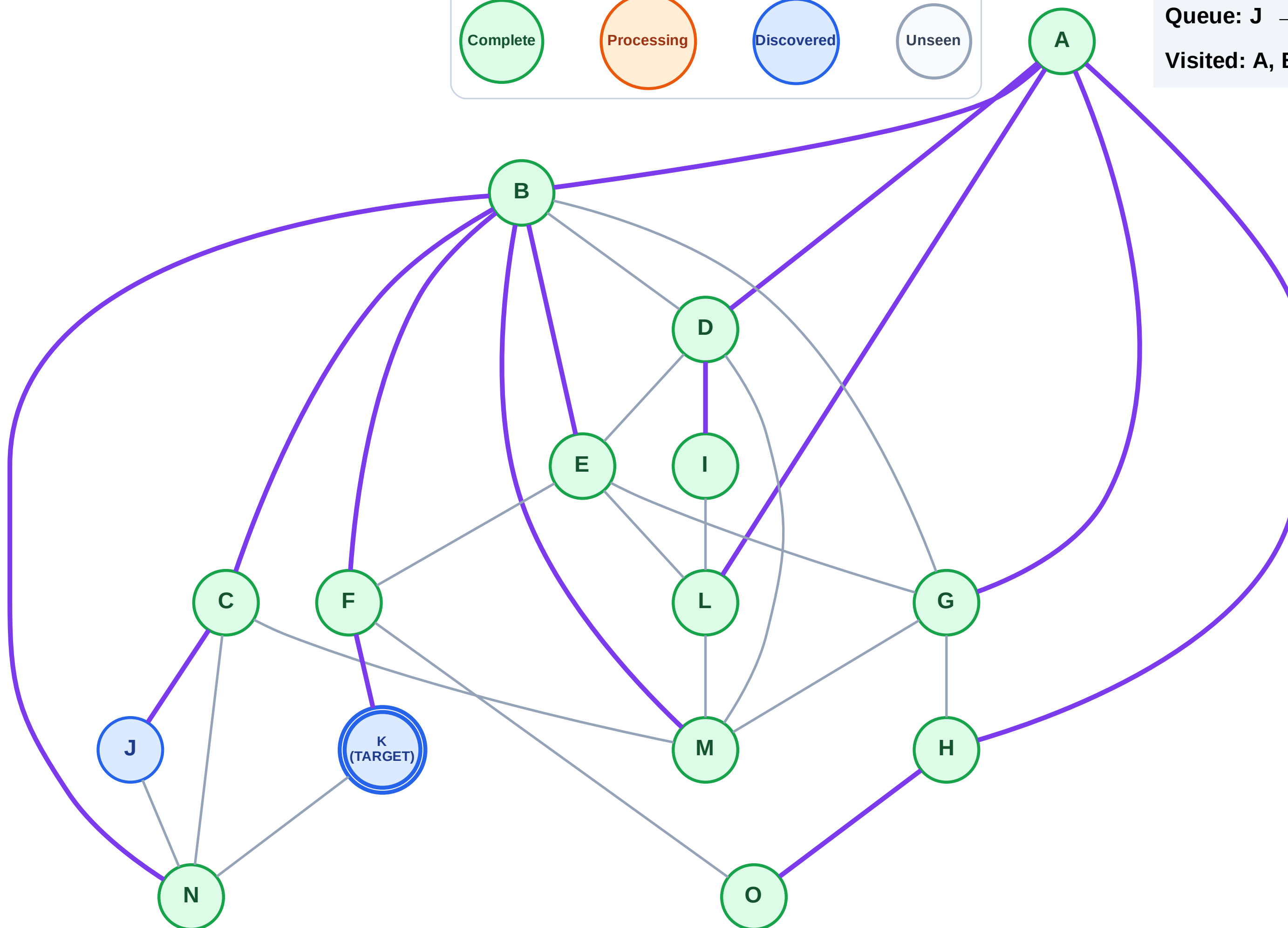
Discovered



Unseen

Queue: J → K

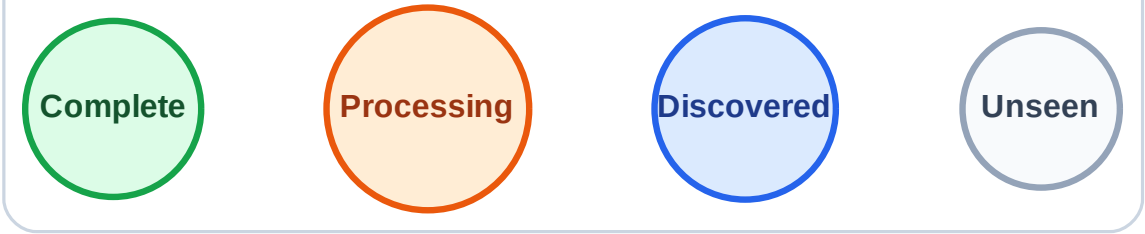
Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

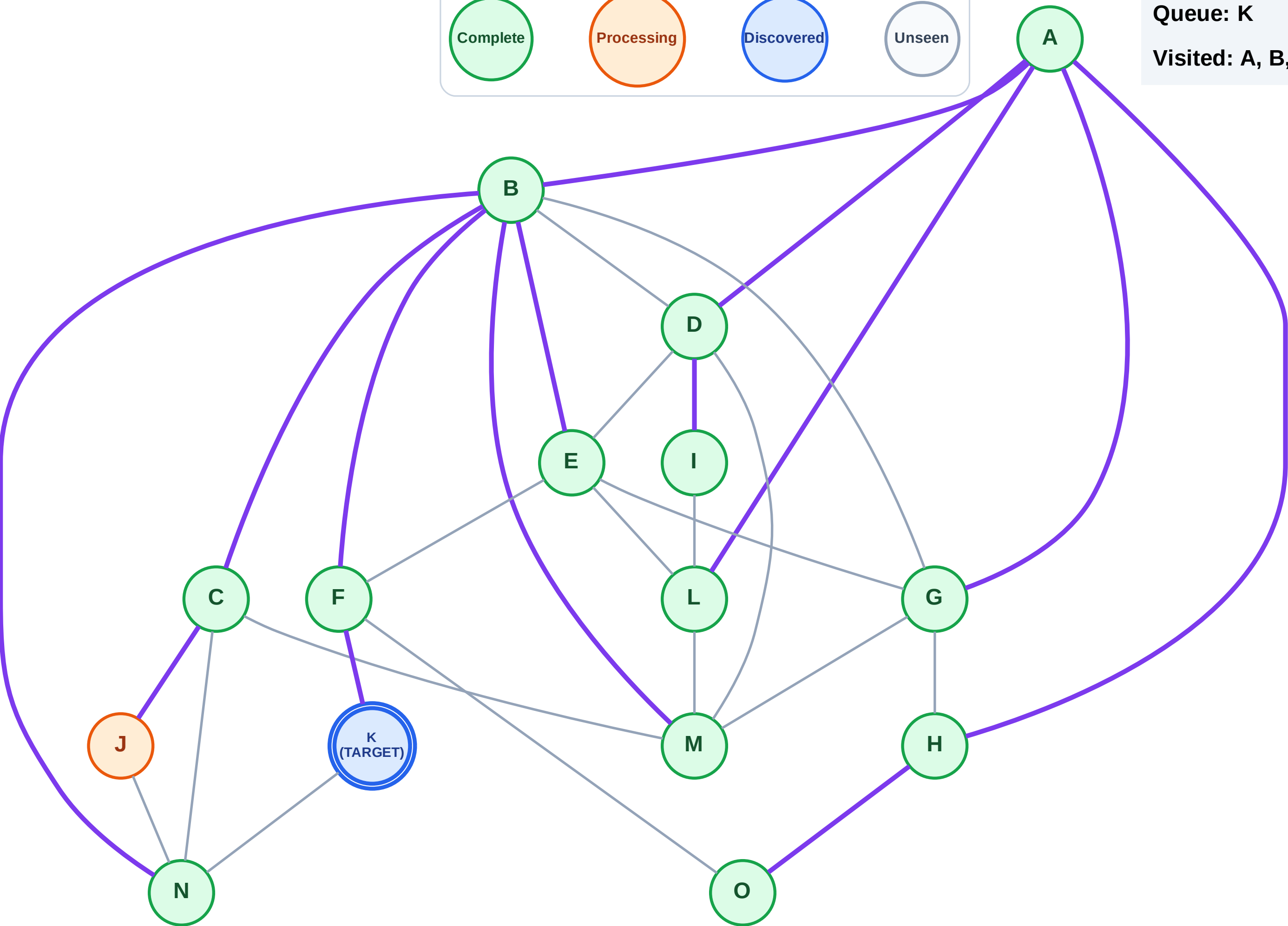
Step 28: Process node J

Legend



Queue: K

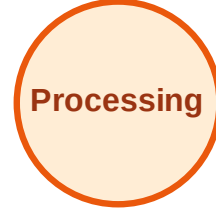
Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

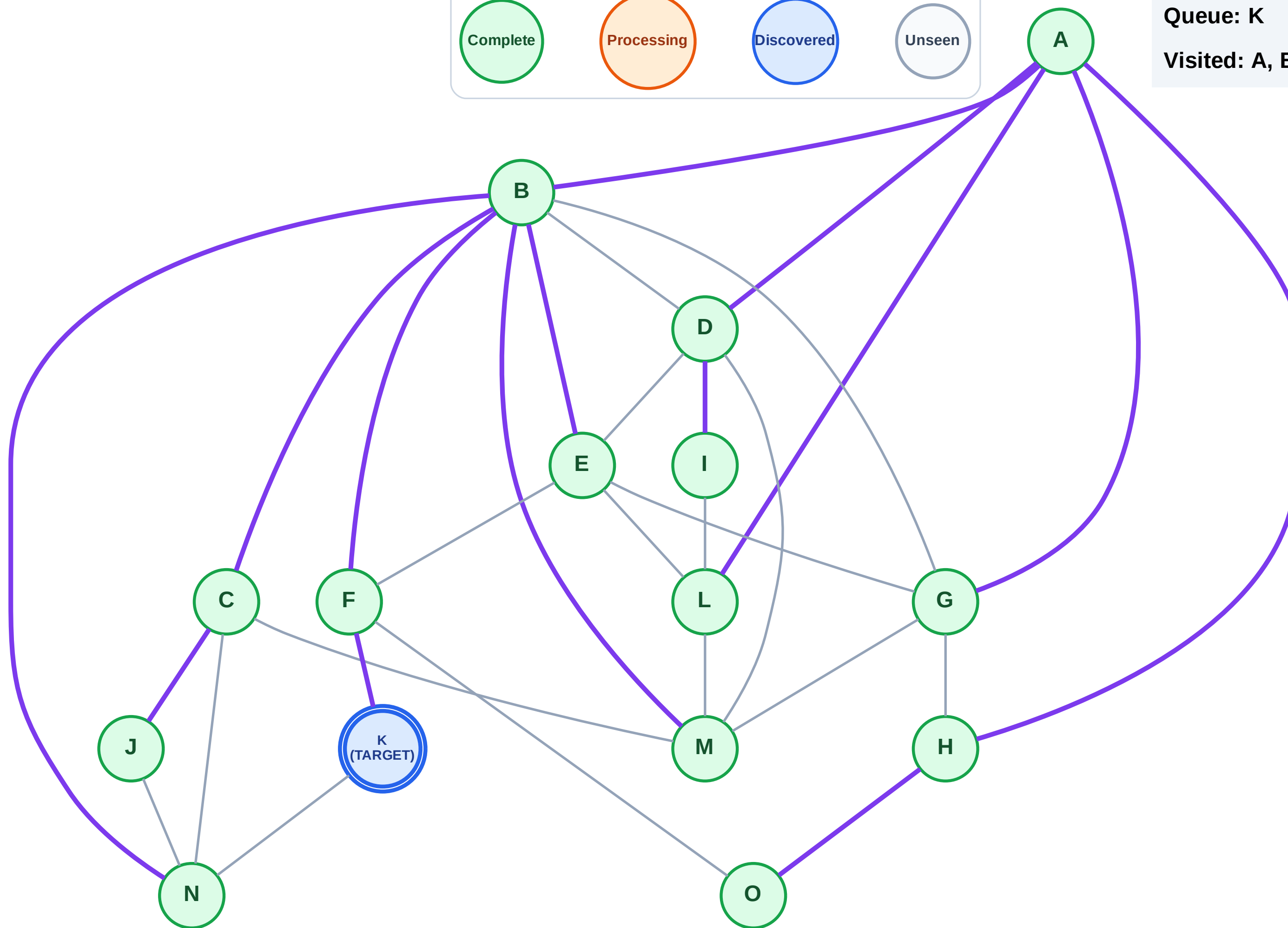
Step 29: No new neighbors from J

Legend



Queue: K

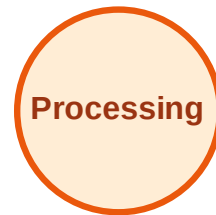
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

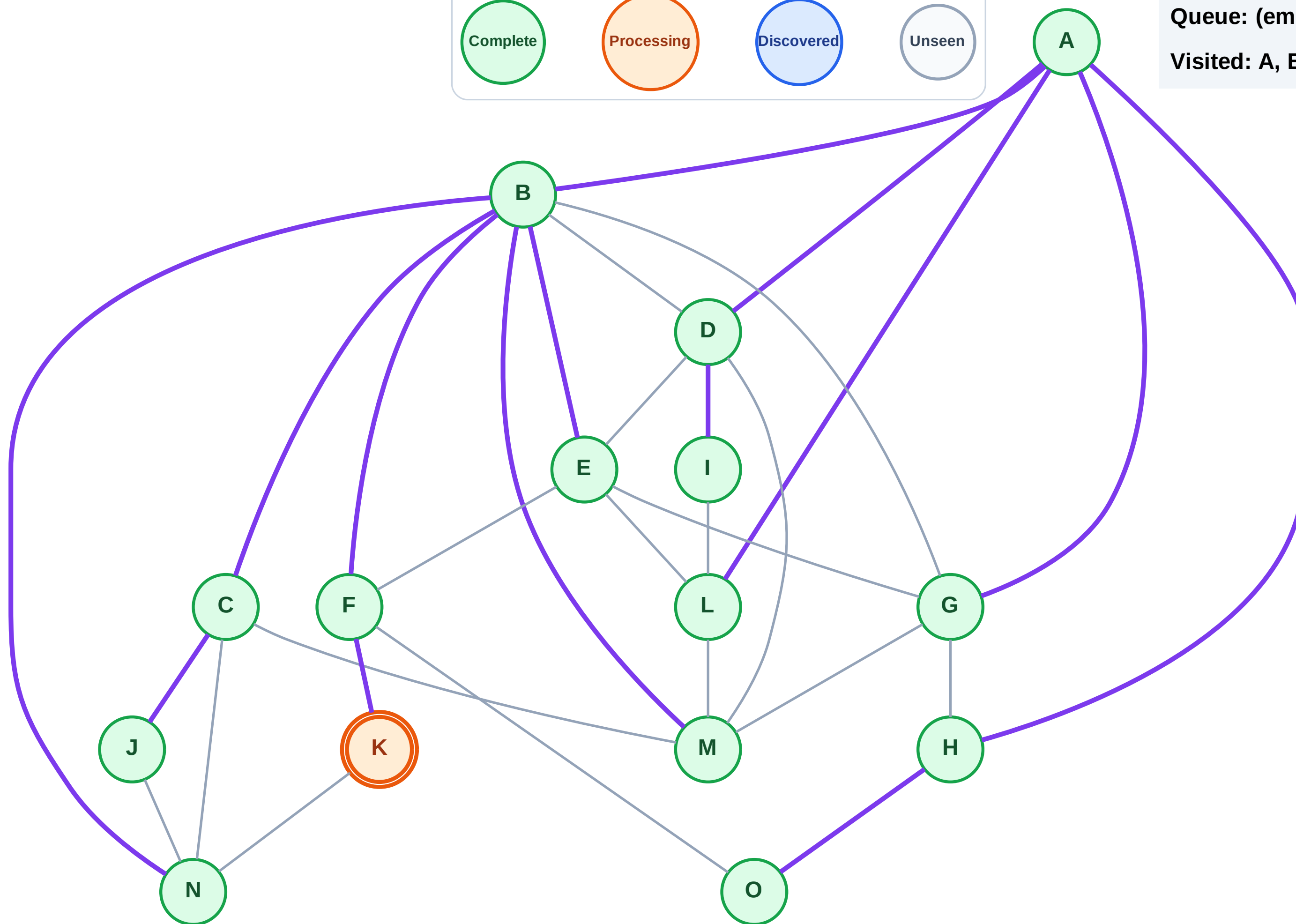
Step 30: Process node K

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



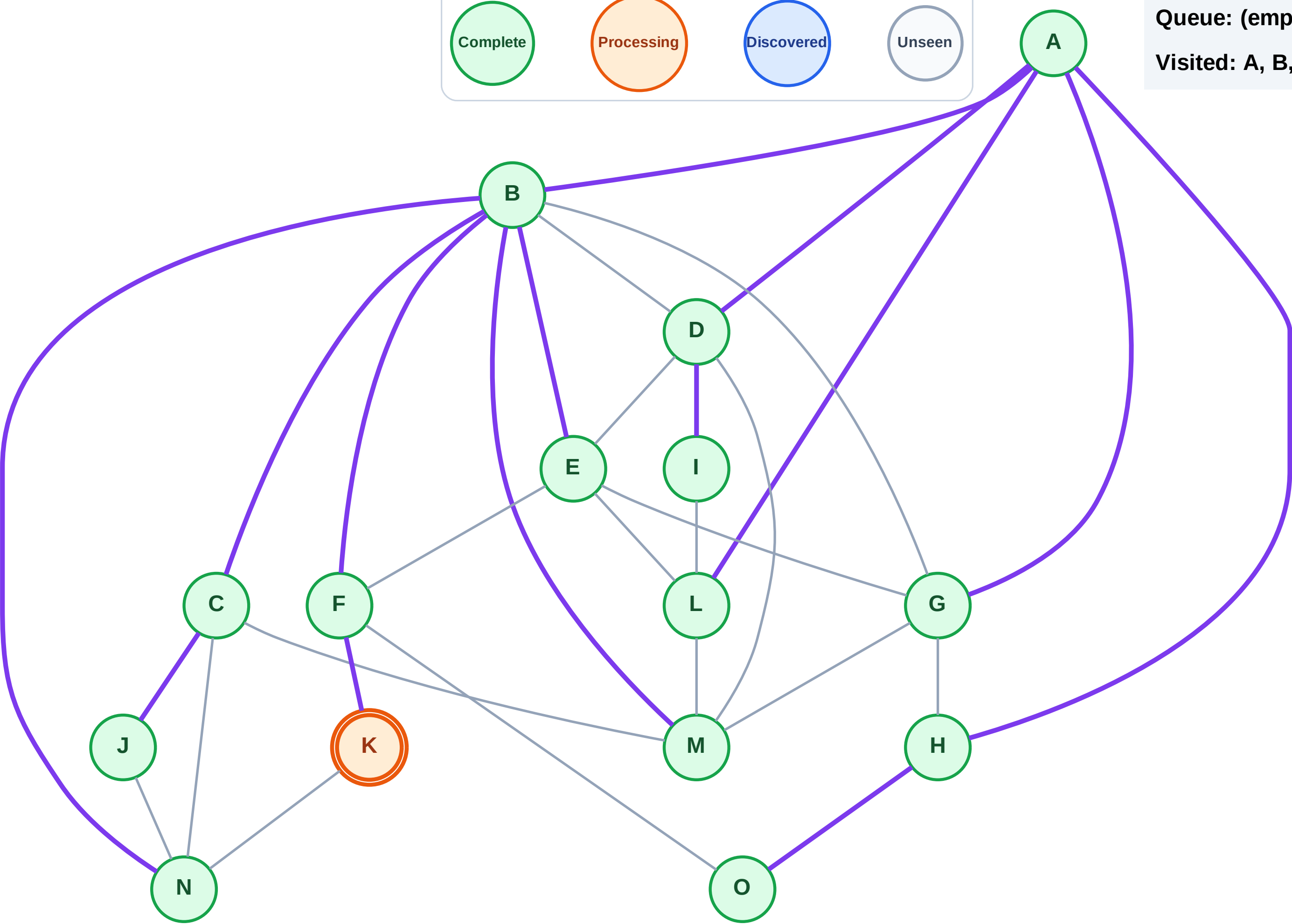
BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend



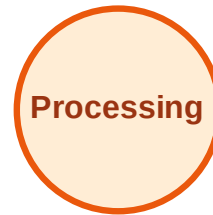
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

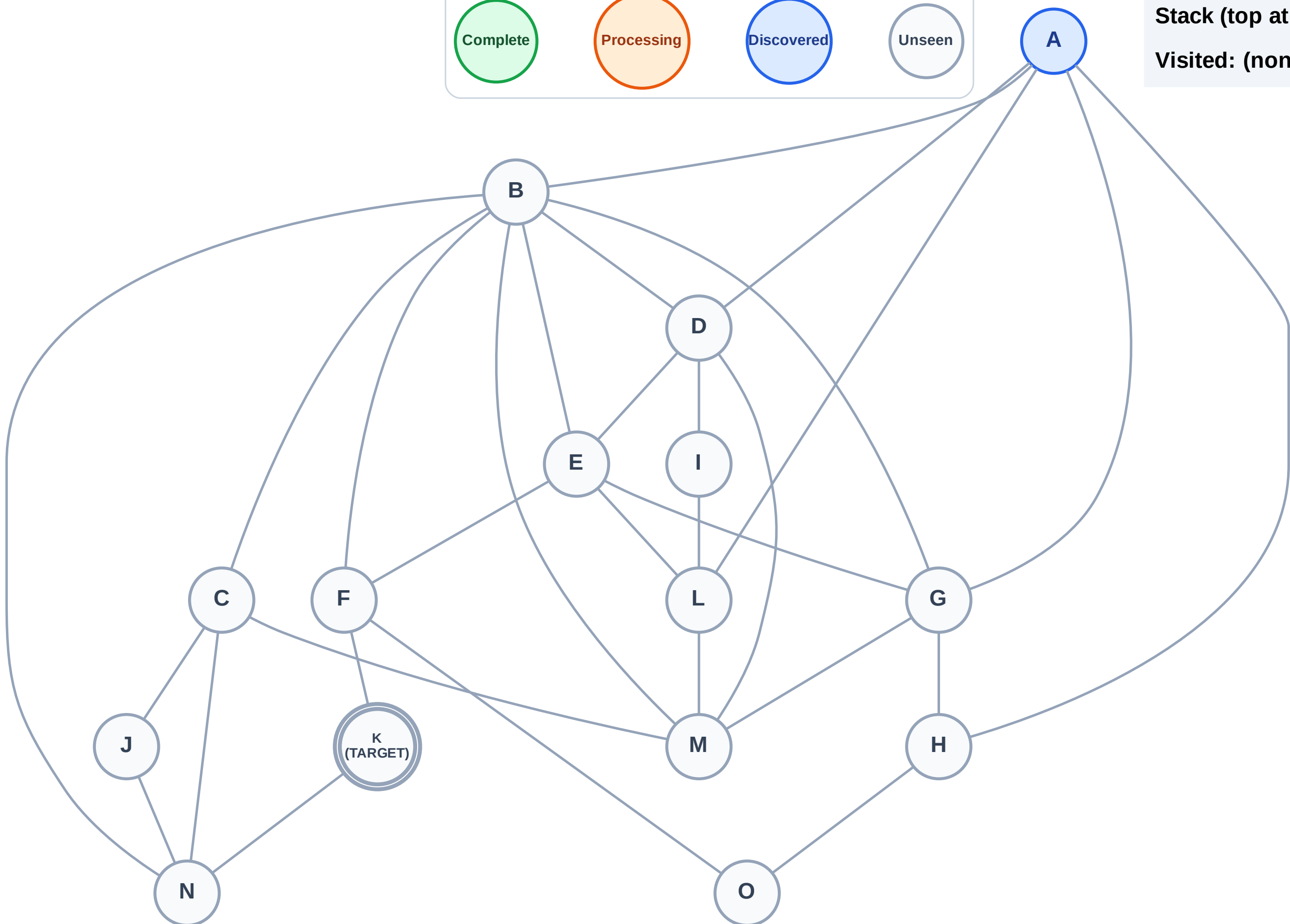
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

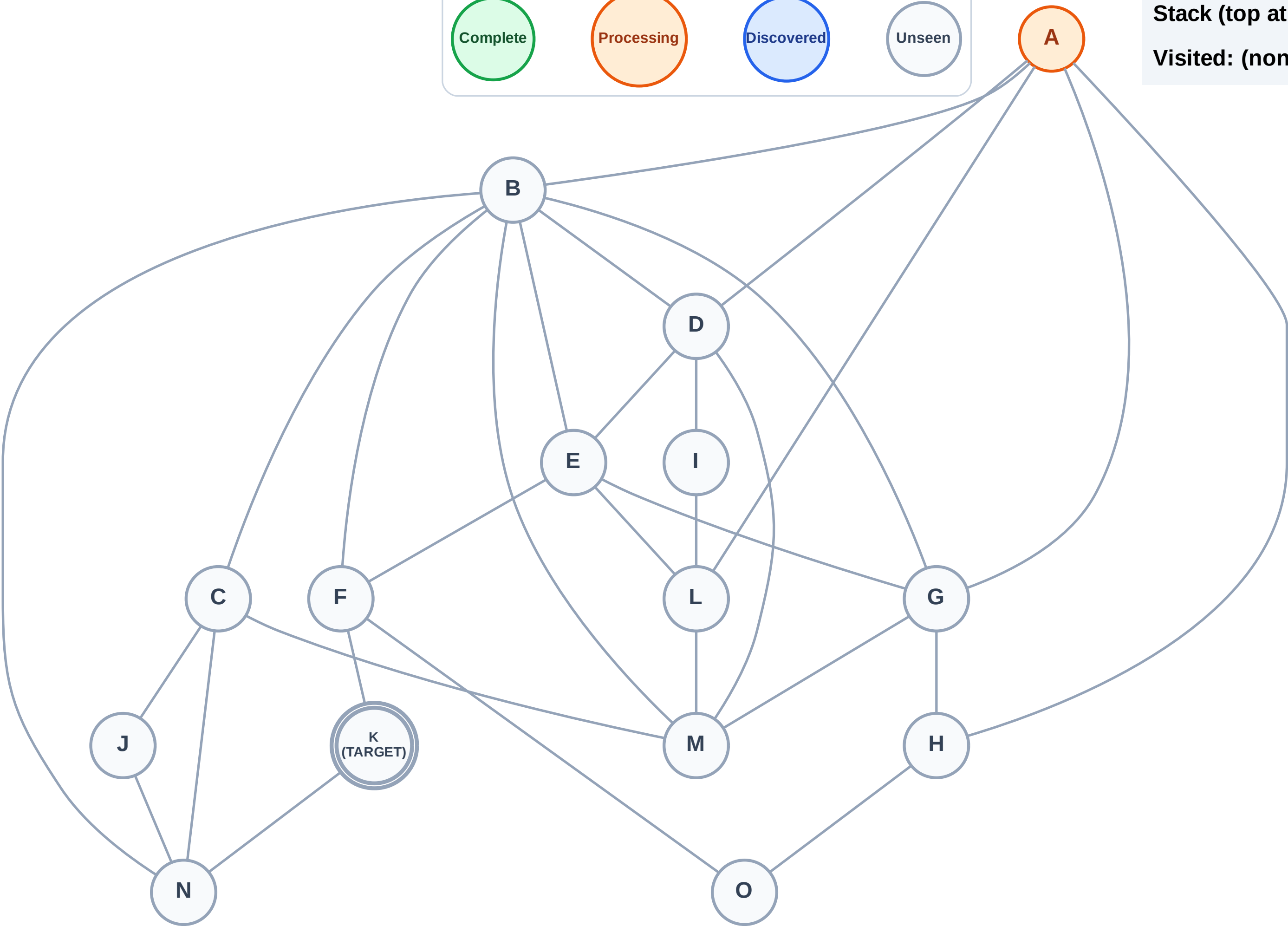
Complete

Processing

Discovered

Unseen

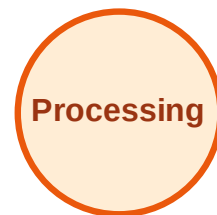
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

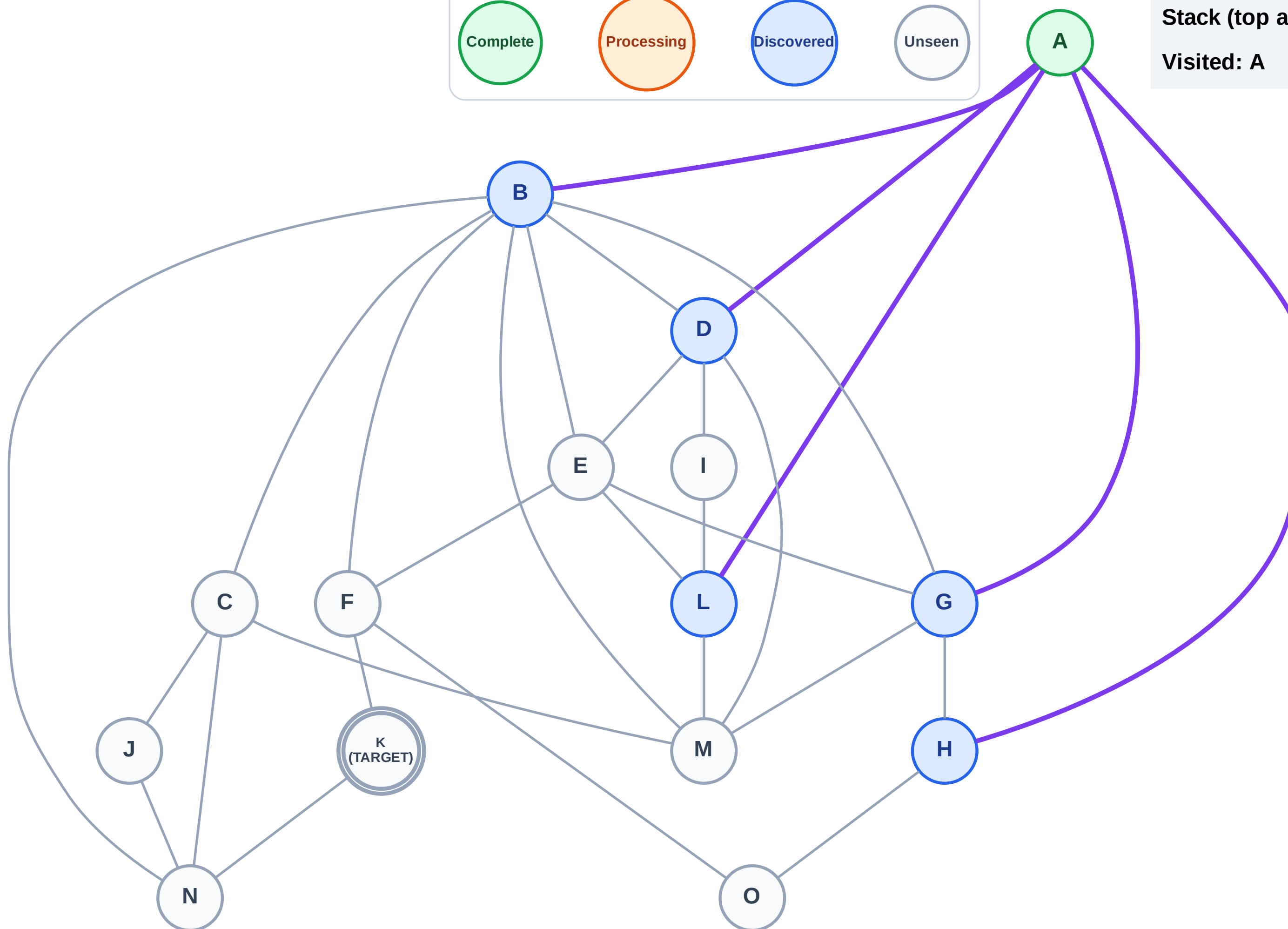
Step 3: Discover L, H, G, D, B from A

Legend



Stack (top at right): L | H | G | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

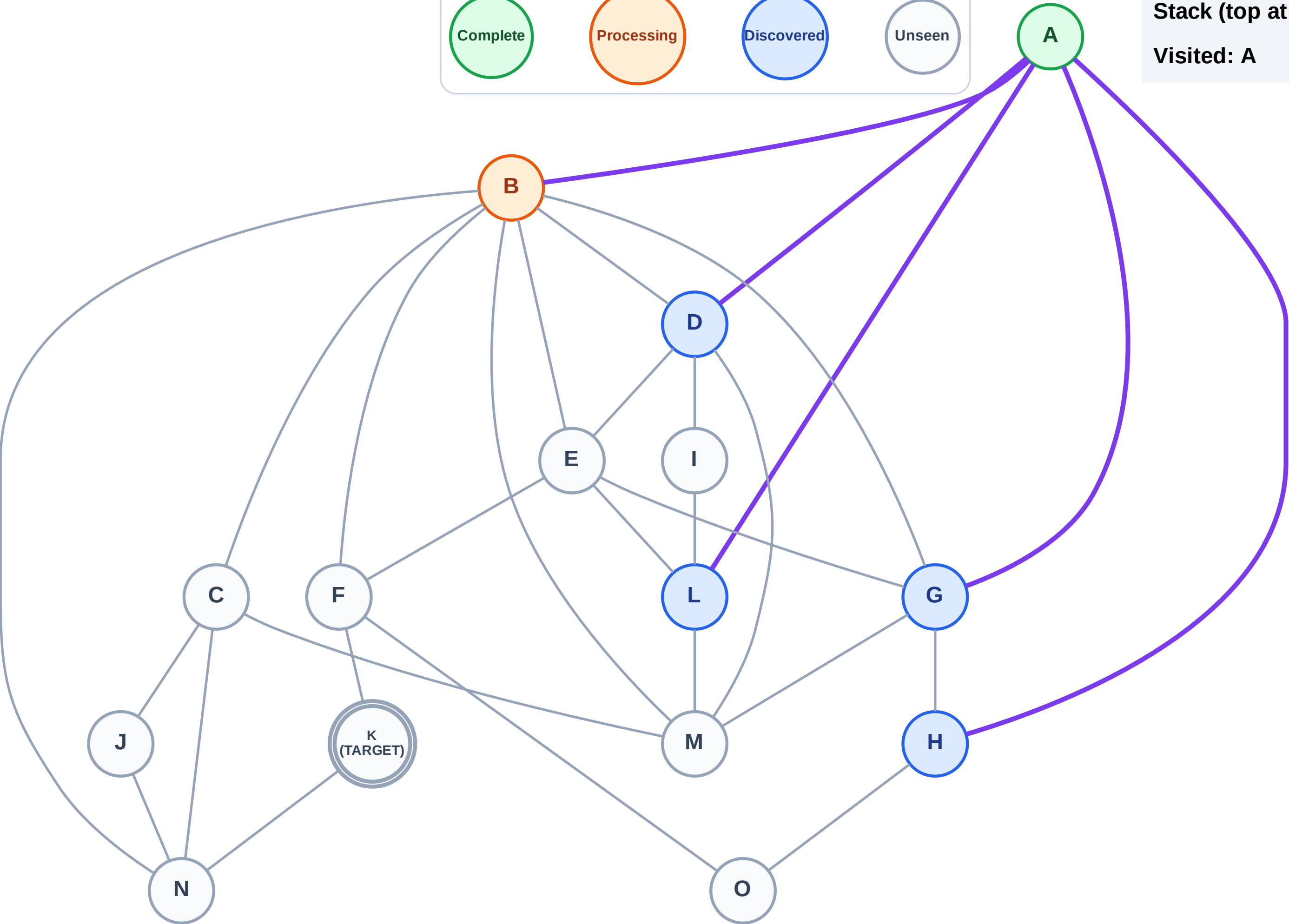
Processing

Discovered

Unseen

Stack (top at right): L | H | G | D

Visited: A



DFS Traversal

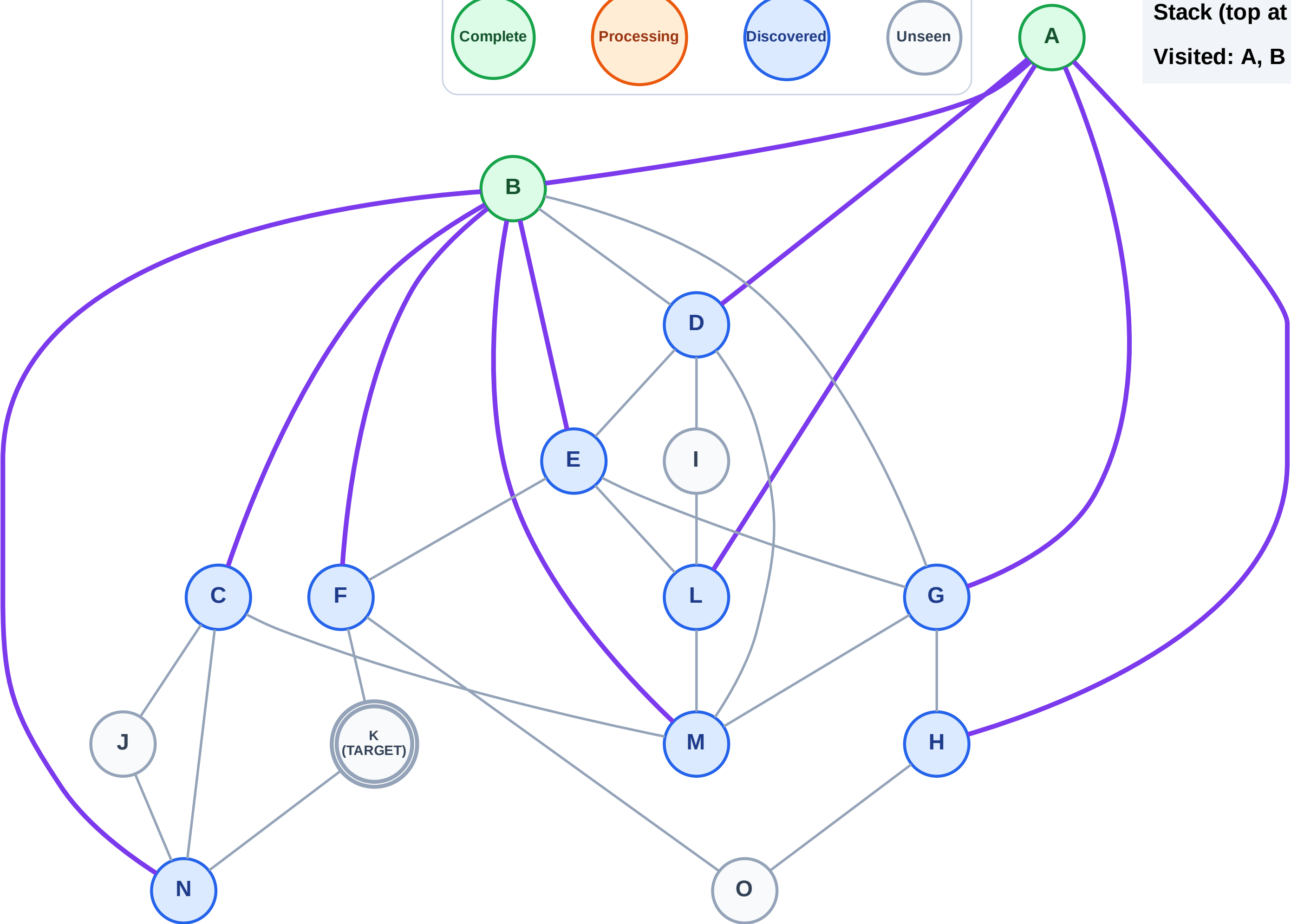
Step 5: Discover N, M, F, E, C from B

Legend



Stack (top at right): L | H | G | D | N | M | F | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

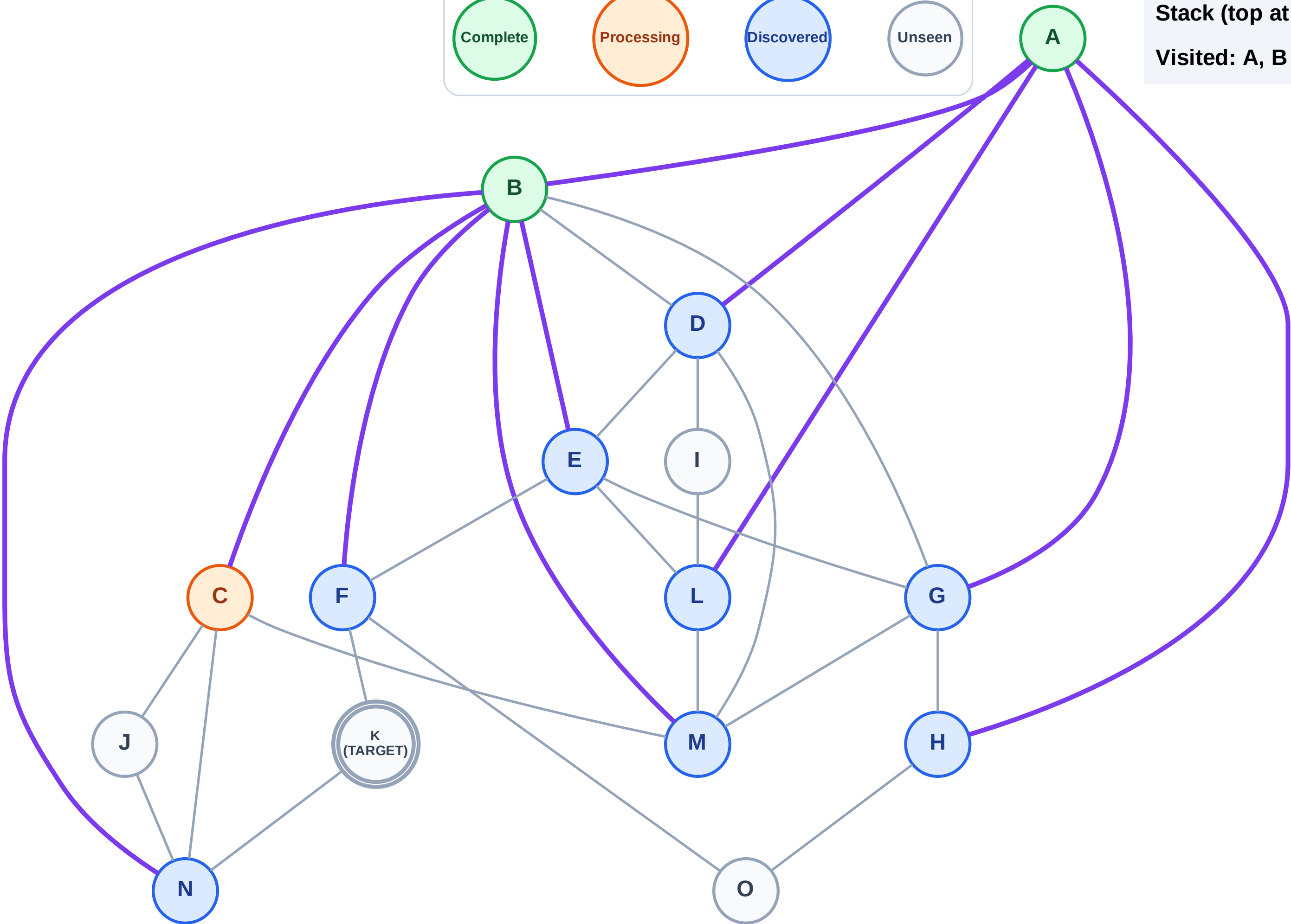
Processing

Discovered

Unseen

Stack (top at right): L | H | G | D | N | M | F | E

Visited: A, B



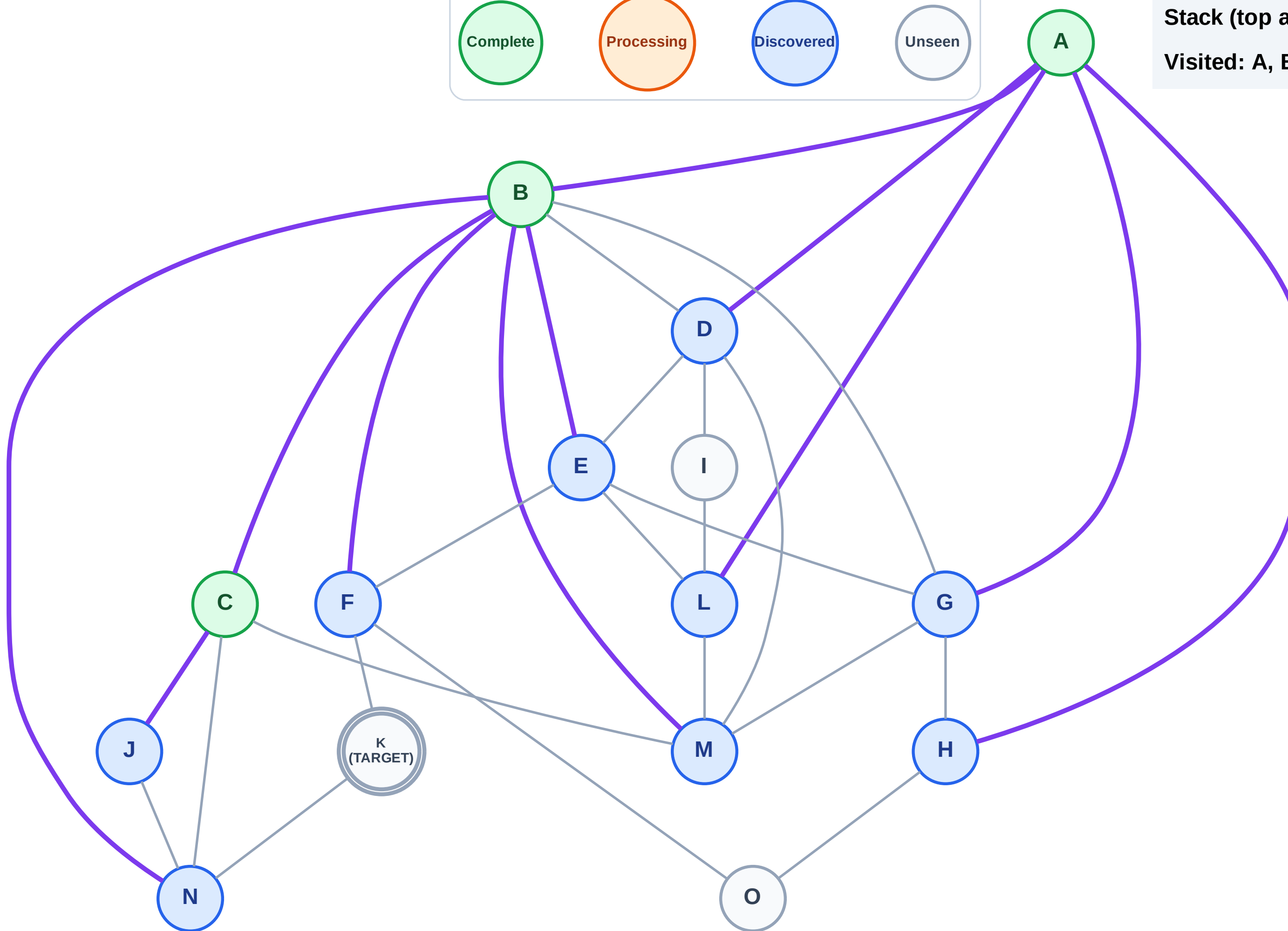
DFS Traversal

Step 7: Discover J from C

Legend



Stack (top at right): L | H | G | D | N | M | F | E | J
Visited: A, B, C



DFS Traversal

Step 8: Process node J

Legend

Complete

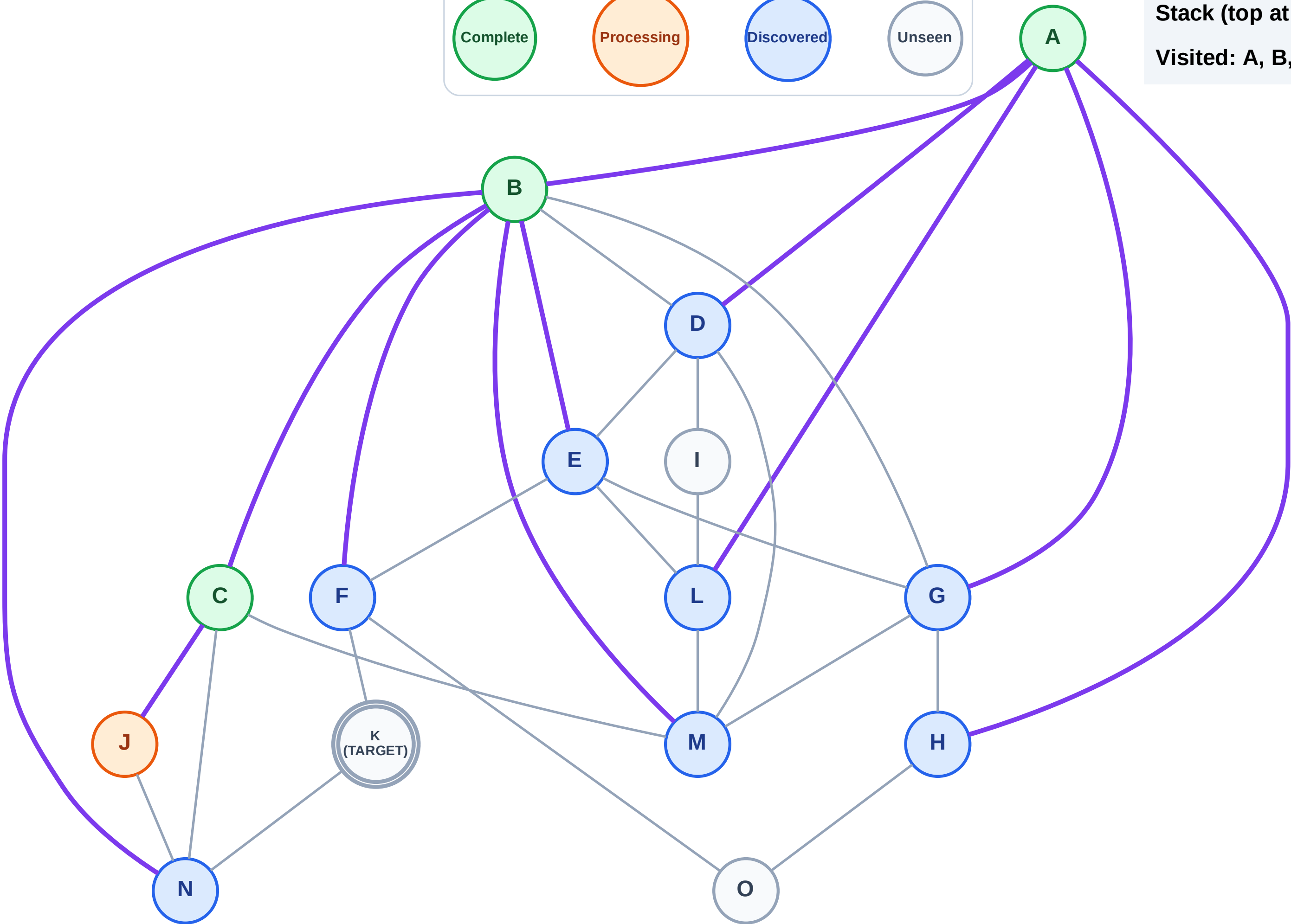
Processing

Discovered

Unseen

Stack (top at right): L | H | G | D | N | M | F | E

Visited: A, B, C



DFS Traversal

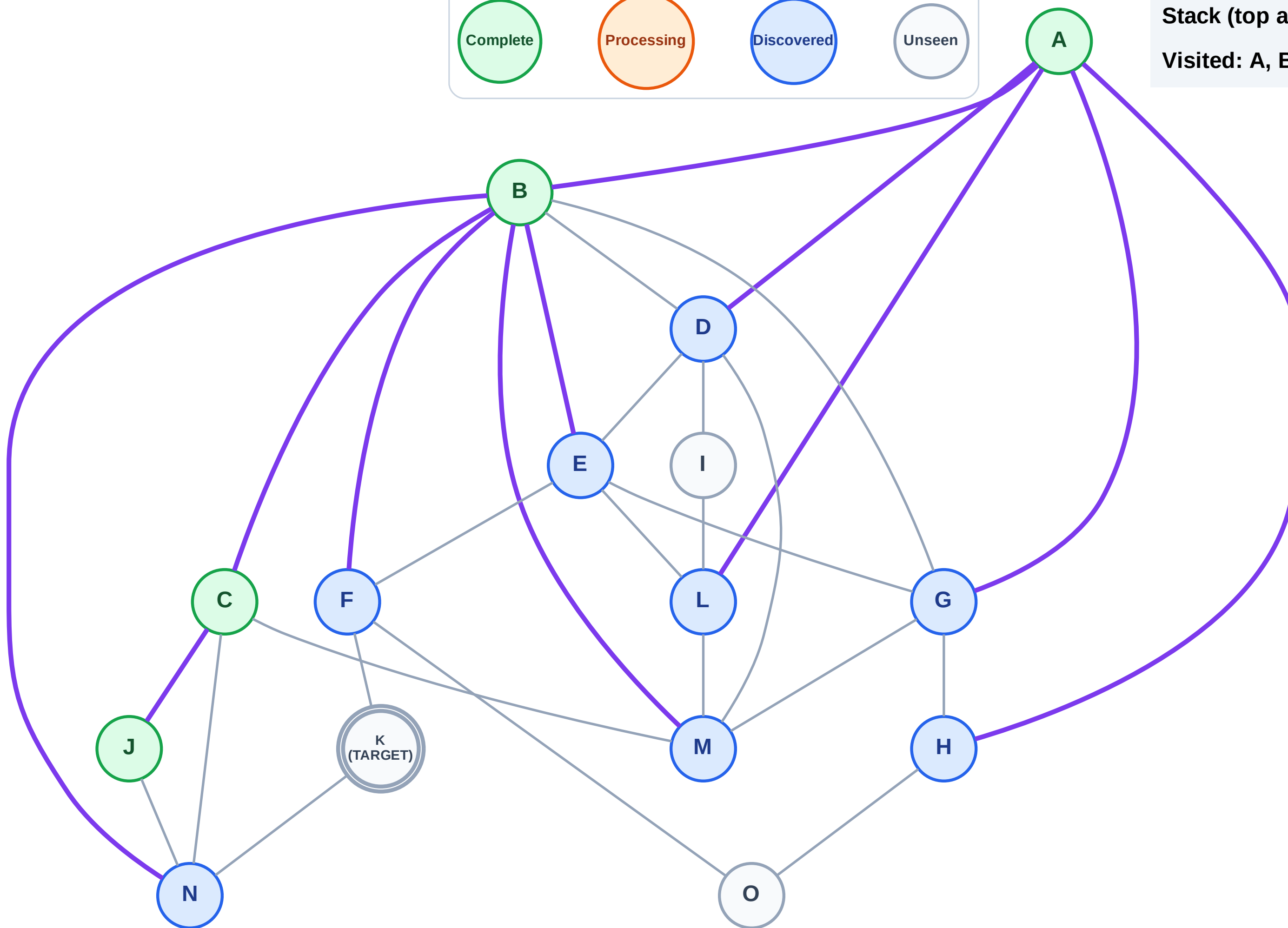
Step 9: No new neighbors from J

Legend



Stack (top at right): L | H | G | D | N | M | F | E

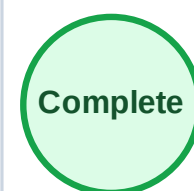
Visited: A, B, C, J



DFS Traversal

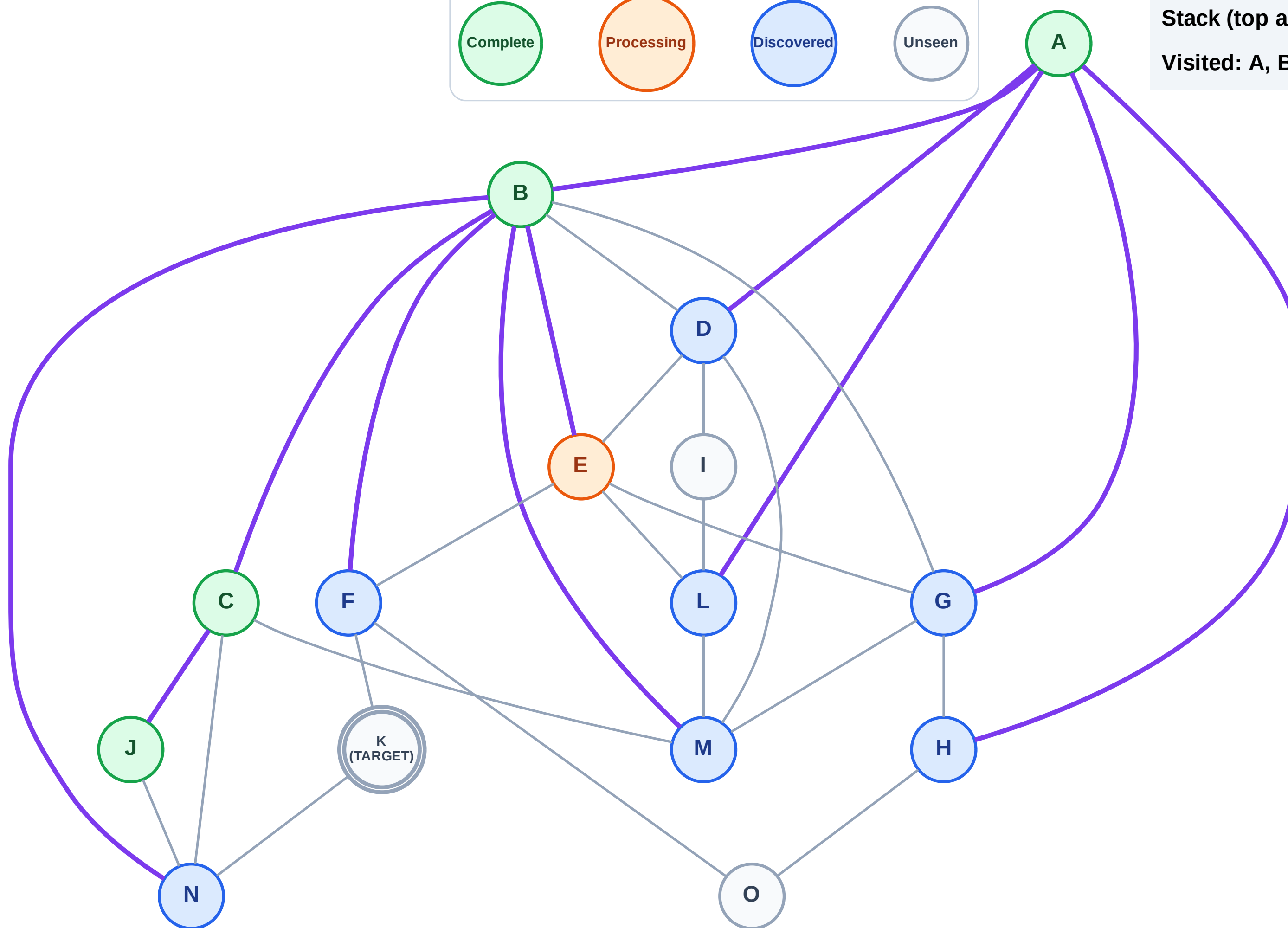
Step 10: Process node E

Legend



Stack (top at right): L | H | G | D | N | M | F

Visited: A, B, C, J



DFS Traversal

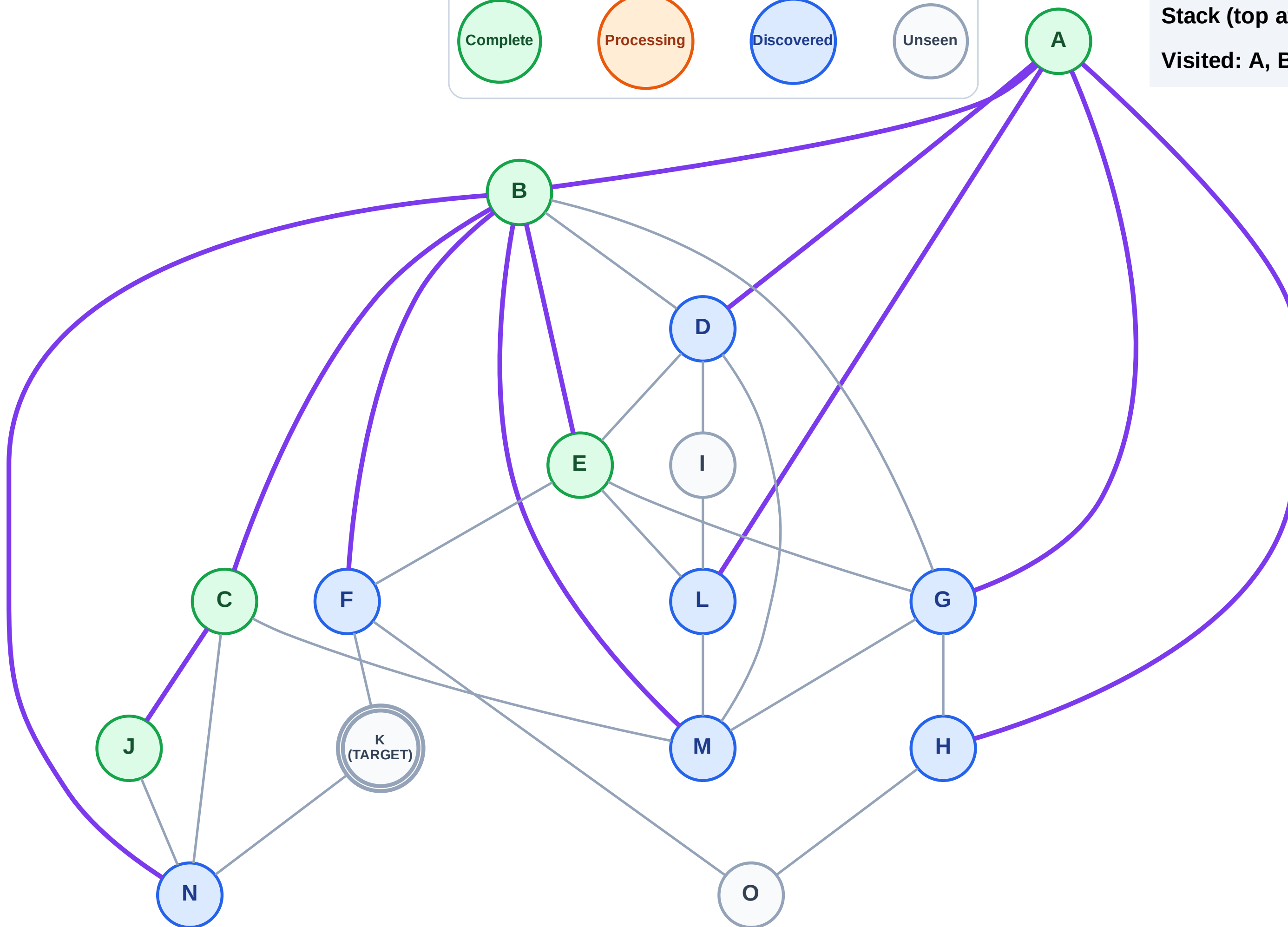
Step 11: No new neighbors from E

Legend



Stack (top at right): L | H | G | D | N | M | F

Visited: A, B, C, E, J



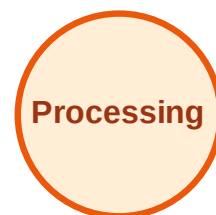
DFS Traversal

Step 12: Process node F

Legend



Complete



Processing



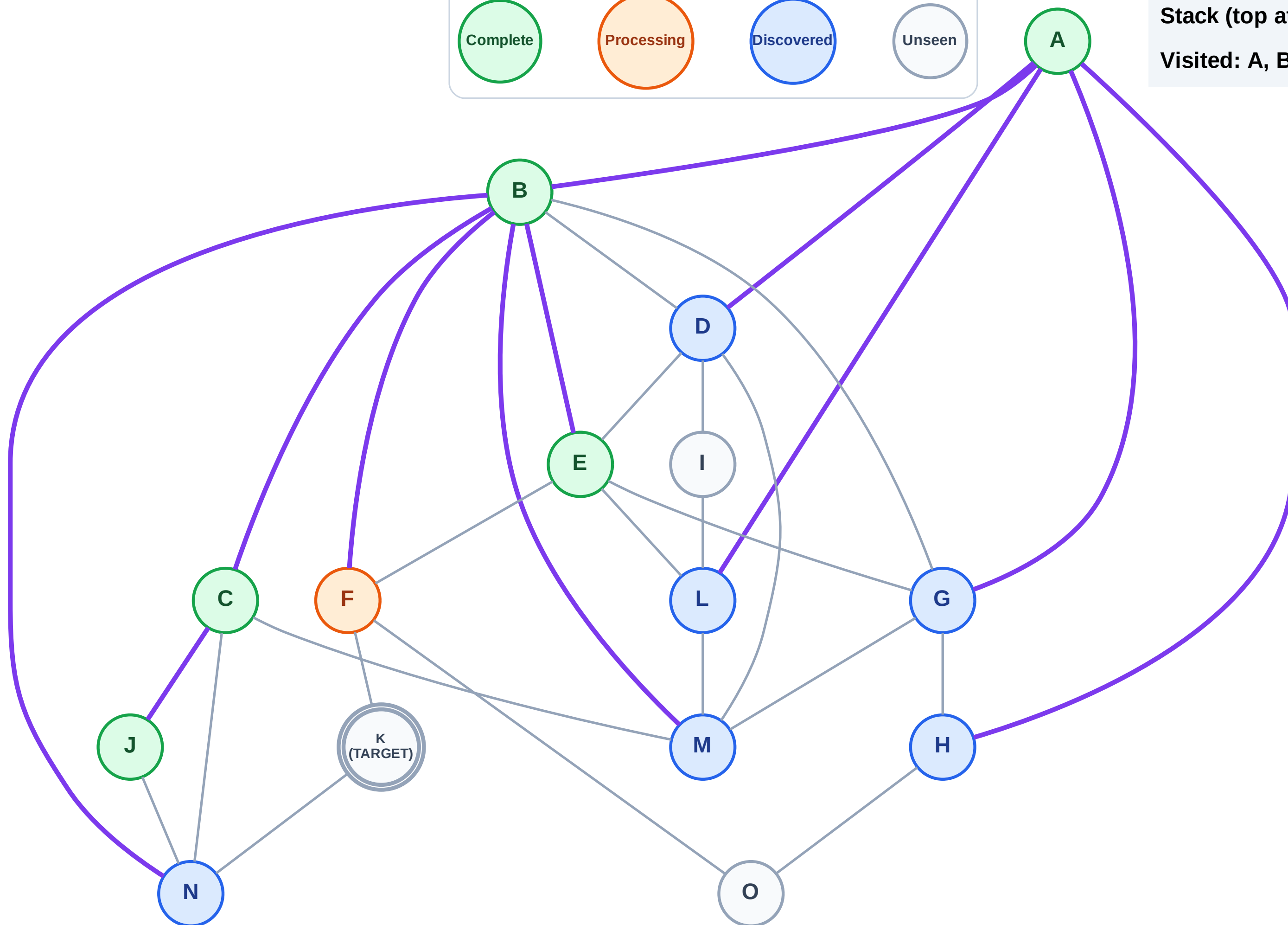
Discovered



Unseen

Stack (top at right): L | H | G | D | N | M

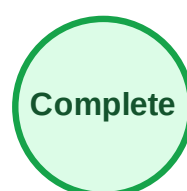
Visited: A, B, C, E, J



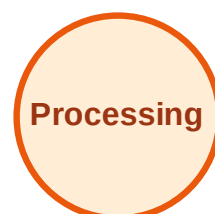
DFS Traversal

Step 13: Discover O, K from F

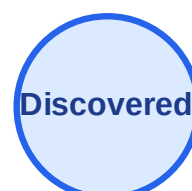
Legend



Complete



Processing



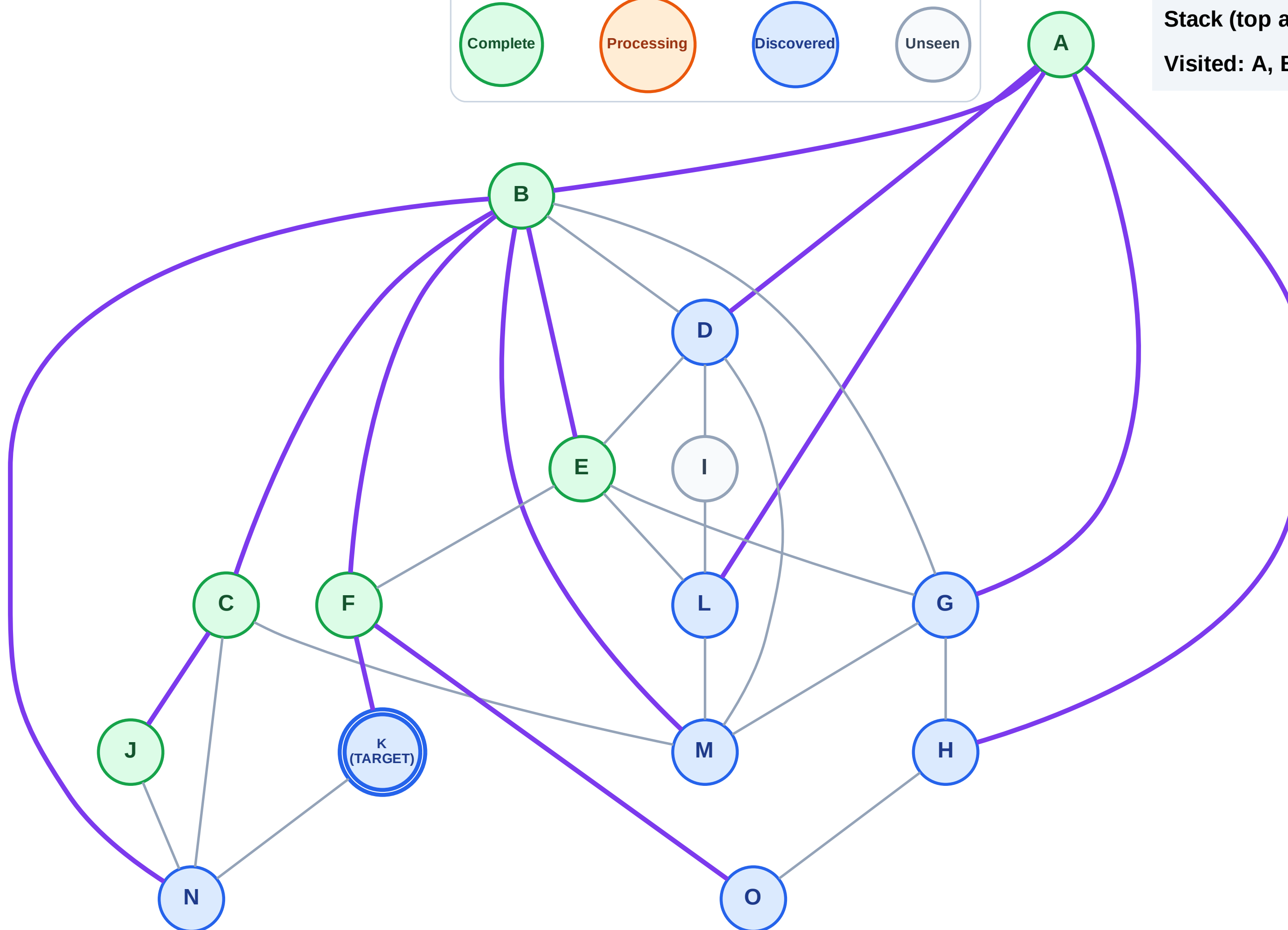
Discovered



Unseen

Stack (top at right): L | H | G | D | N | M | O | K

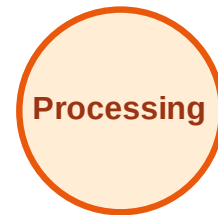
Visited: A, B, C, E, F, J



DFS Traversal

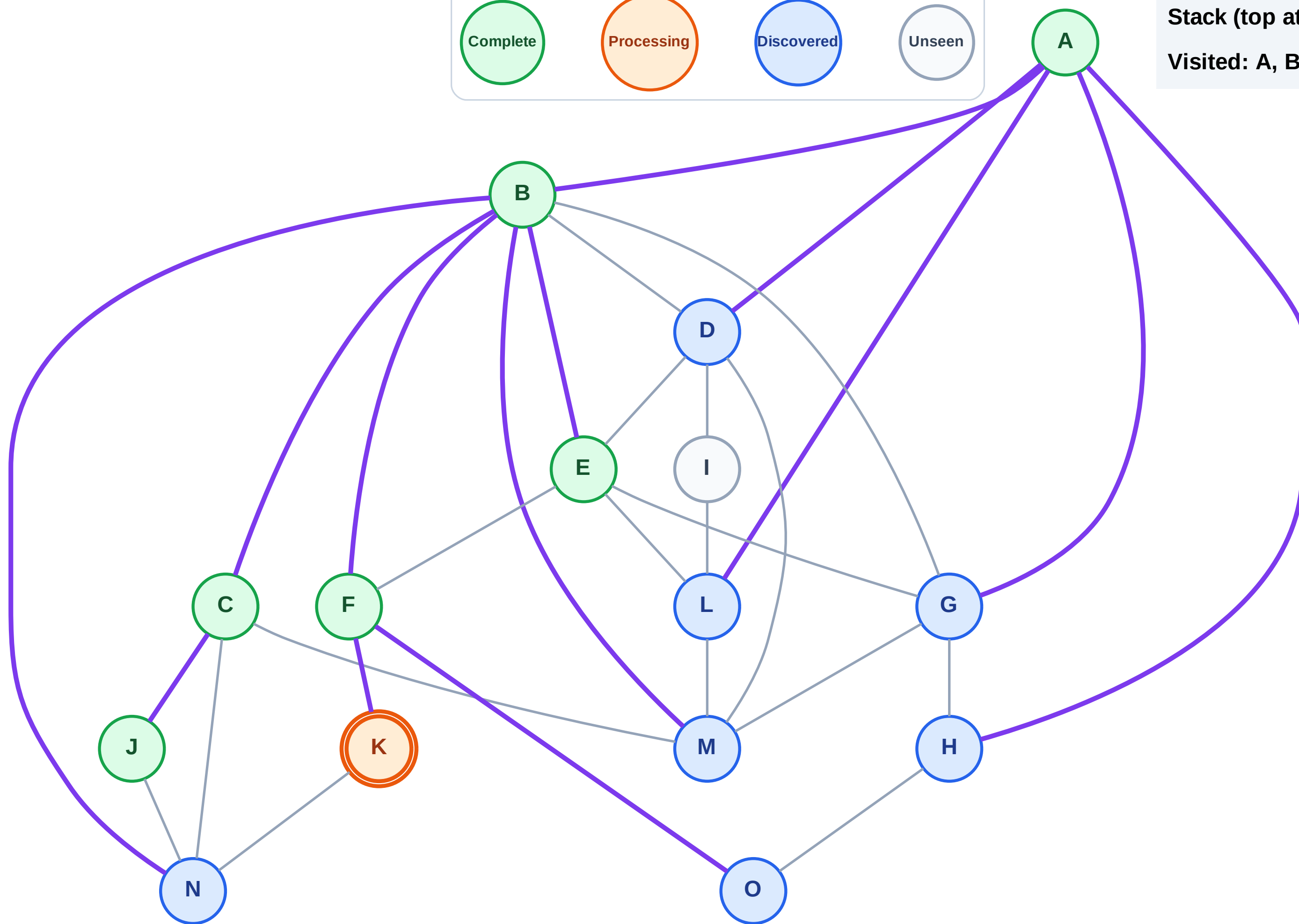
Step 14: Process node K

Legend



Stack (top at right): L | H | G | D | N | M | O

Visited: A, B, C, E, F, J



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | H | G | D | N | M | O

Visited: A, B, C, E, F, J, K

